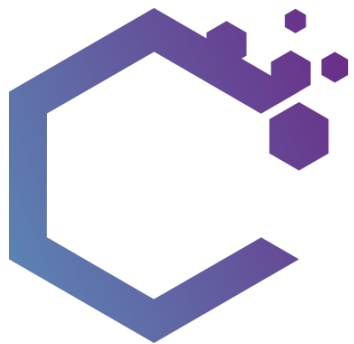




CEISAM

Chimie et Interdisciplinarité
Synthèse, Analyse,
Modélisation



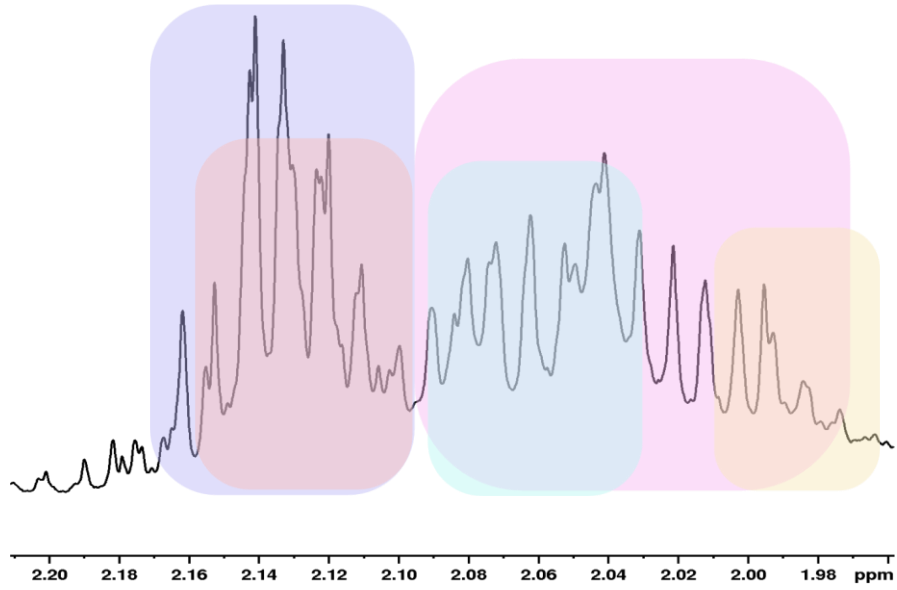
Nantes
Université



Two-dimensional NMR in clinical metabolomics

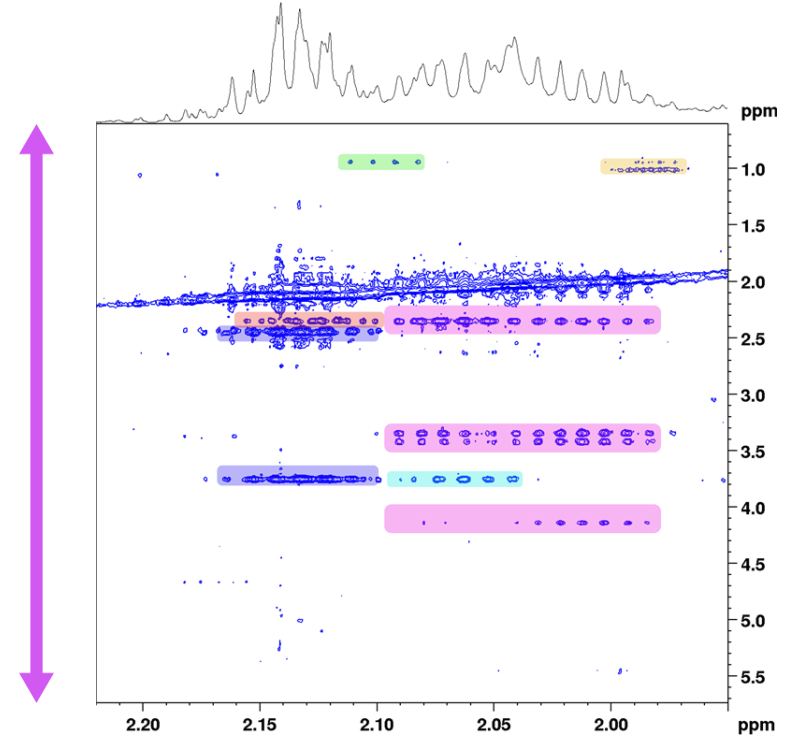
Thematic School #2 – RFMF 07/10/2026

2D NMR in metabolomics



1D ^1H NMR

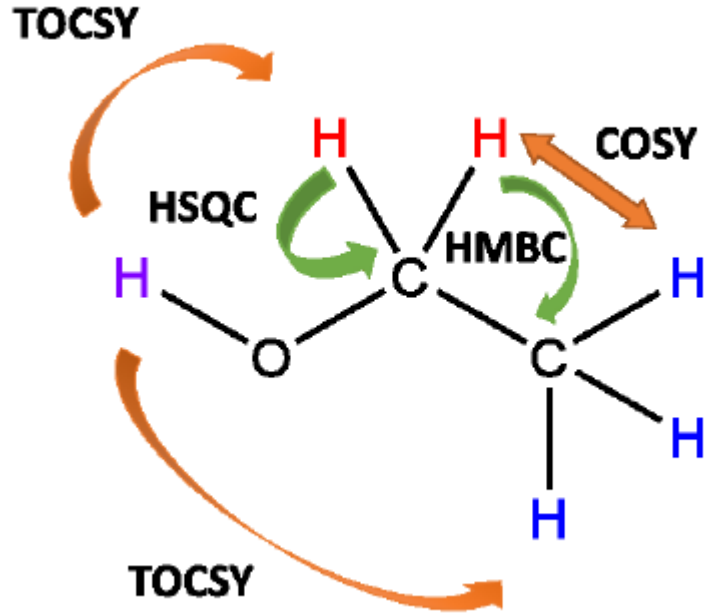
- Short spectral width
- Signal overlapping



2D zTOCSY

- Increased peak separation
- Facilitated bucketing

Multidimensional NMR to improve resolution



Homonuclear experiments (^1H - ^1H)

- J-resolved (JRES)
- Correlation Spectroscopy (COSY)
- Total Correlation Spectroscopy (TOCSY)

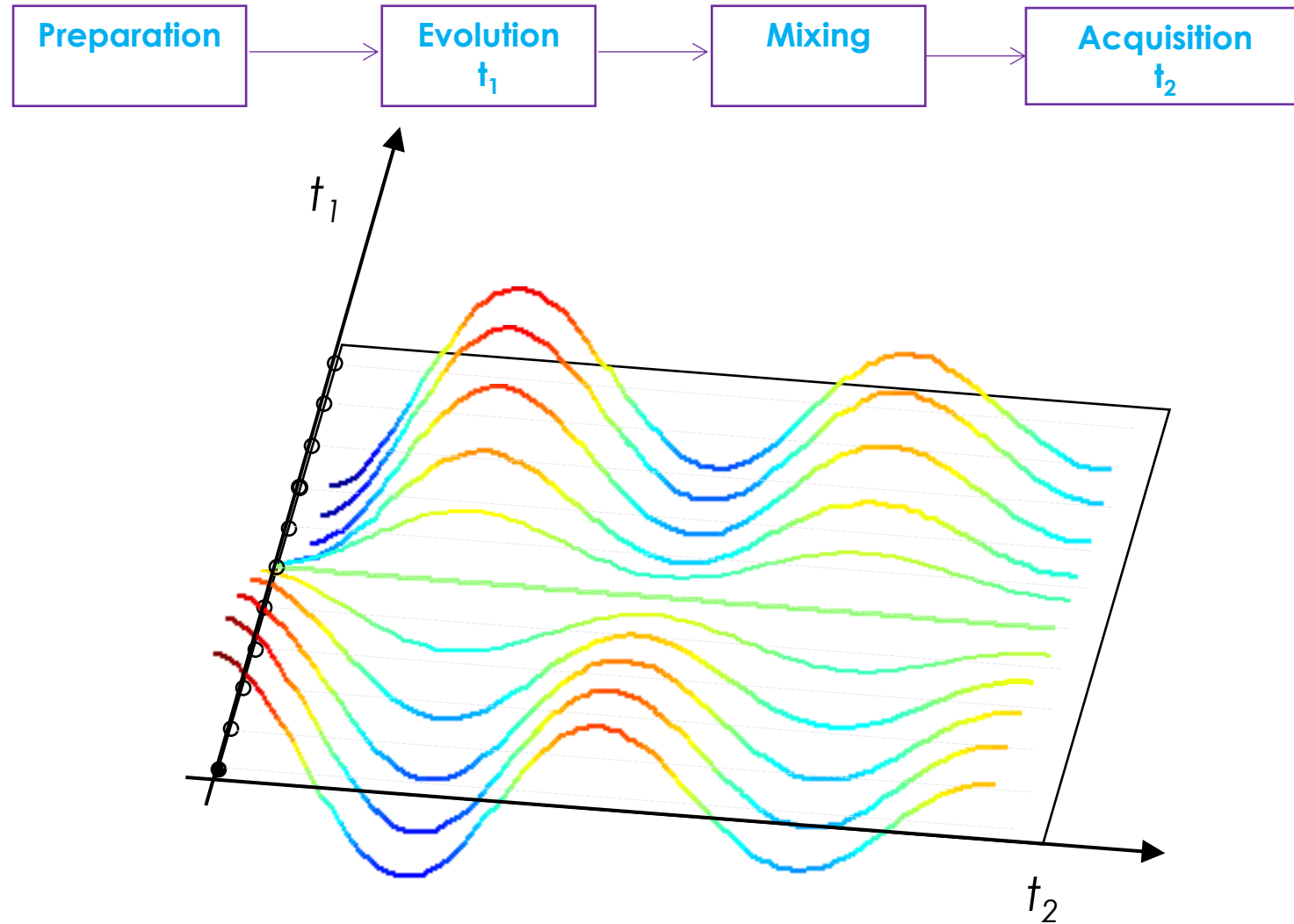
linkage information for J-coupled protons, so J coupling required = *not useful for singlets!*

Heteronuclear experiments (^1H - ^{13}C)

- Heteronuclear Single Quantum Coherence (HSQC)
- Heteronuclear Multiple Bond Correlation (HMBC)

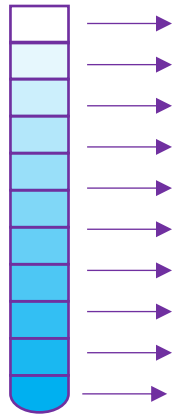
Applied only for structural elucidation, but not systematically in metabolomic workflows.

Experiment time issue



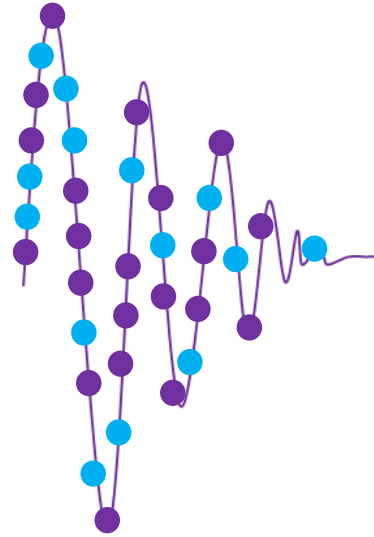
Time saving strategies for 2D NMR metabolomics

UltraFast (UF)



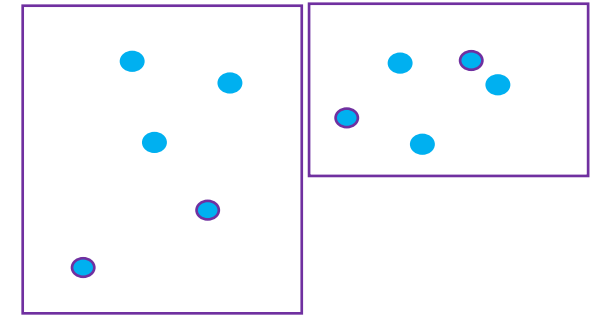
*Simultaneous
acquisition of
different
slices of the
sample*

Non Uniform Sampling (NUS)

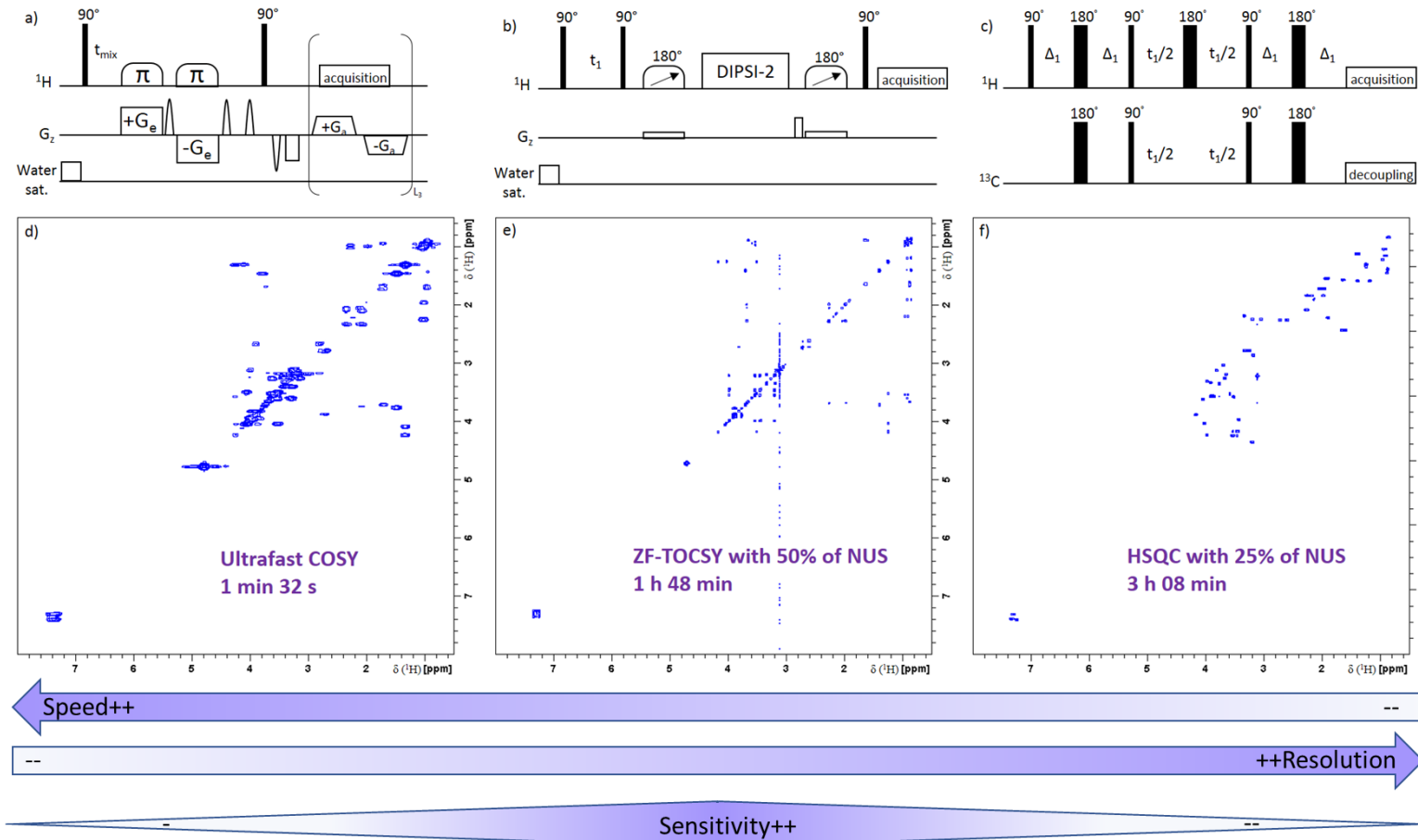


- *Real data*
- *Data reconstructed by an algorithm*

Spectral aliasing

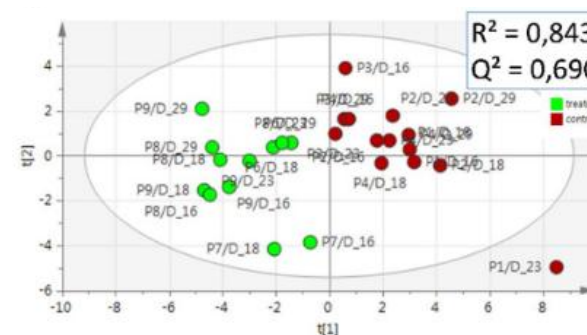
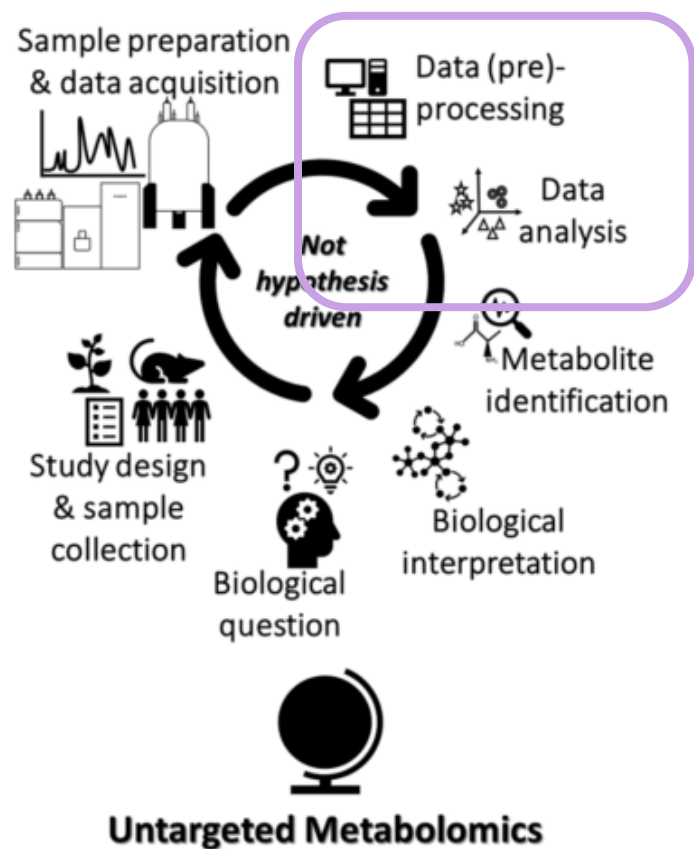


A continuum of fast 2D NMR methods for a better resolution

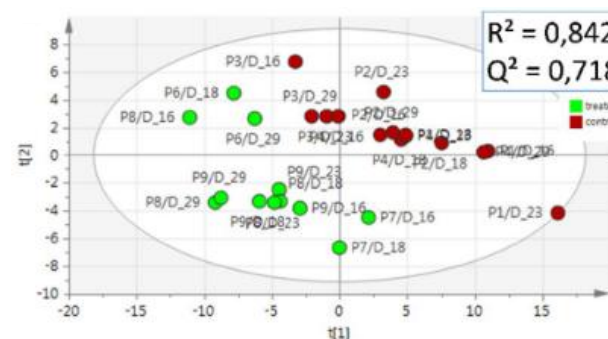


Fast 2D NMR in metabolomics

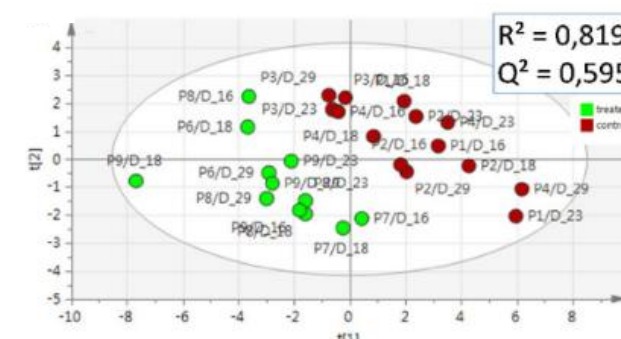
Application to the metabolomics workflow :



1D ^1H NMR



2D NUS zTOCSY

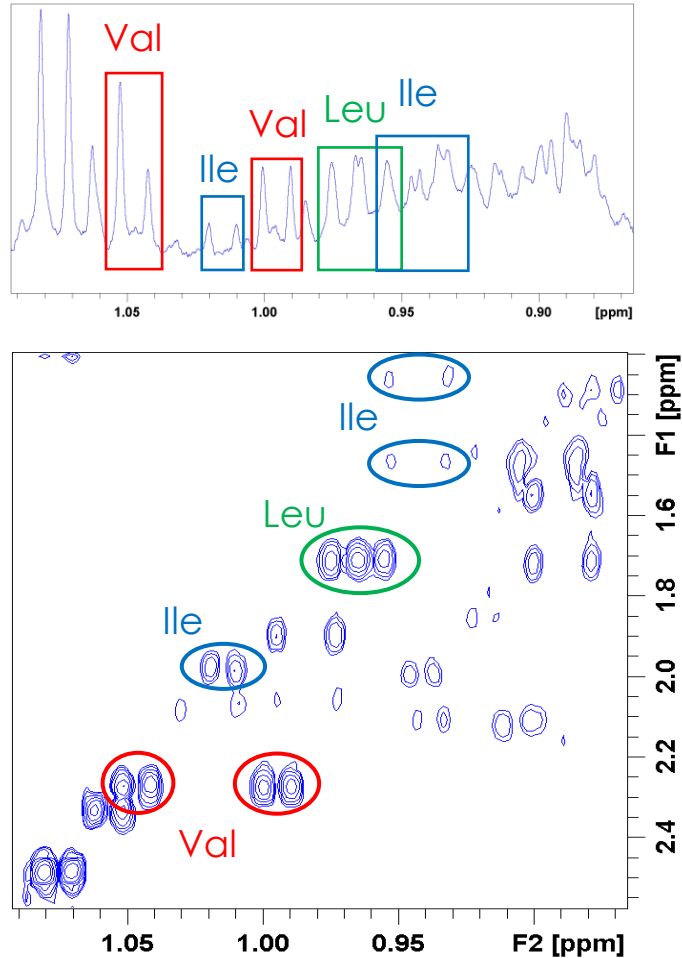


2D UF COSY

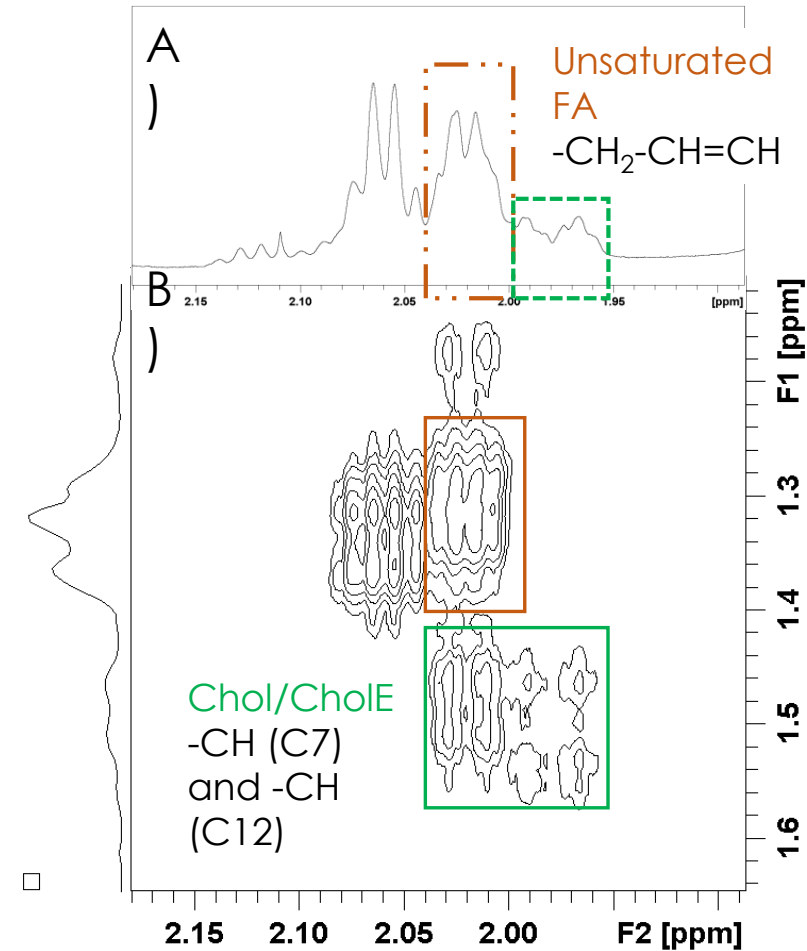
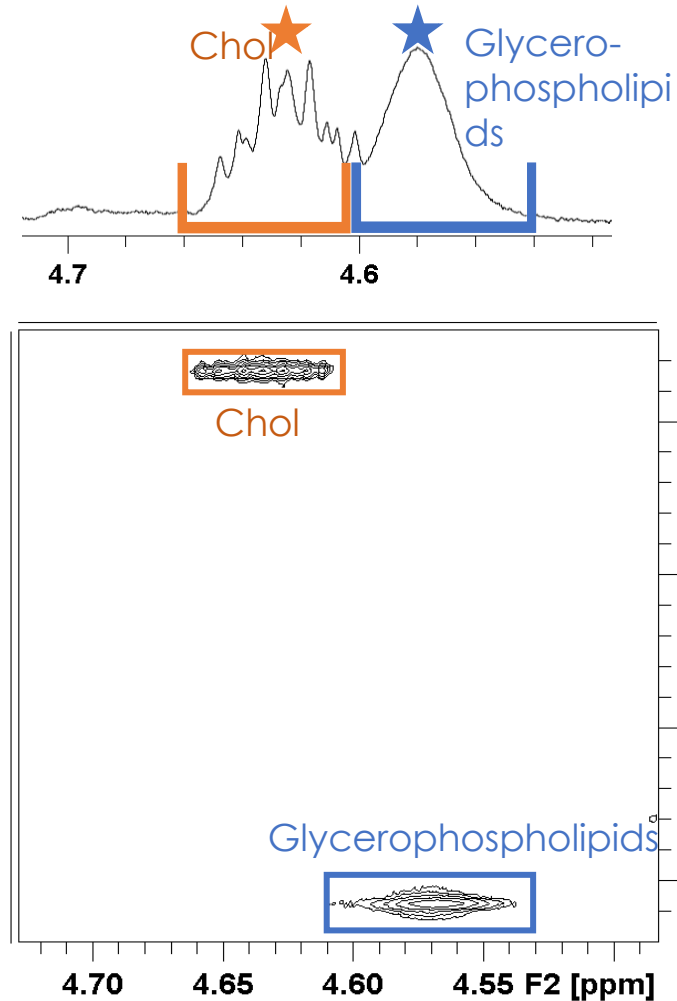
PLS-DA study of lipidomics serum extracts from pigs treated with ractopamine

2D help to overcome peak overlap

In metabolomics:



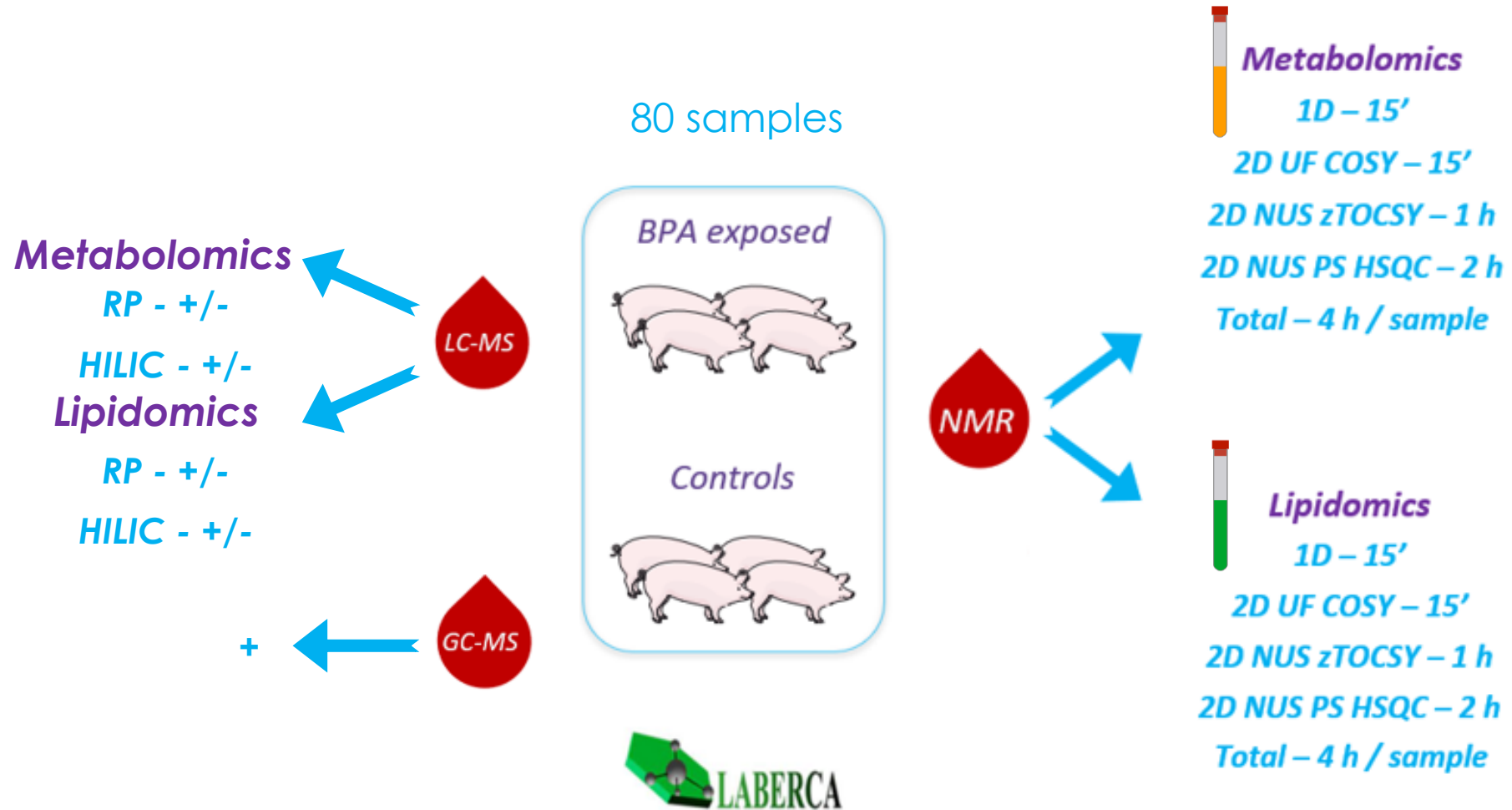
In lipidomics:



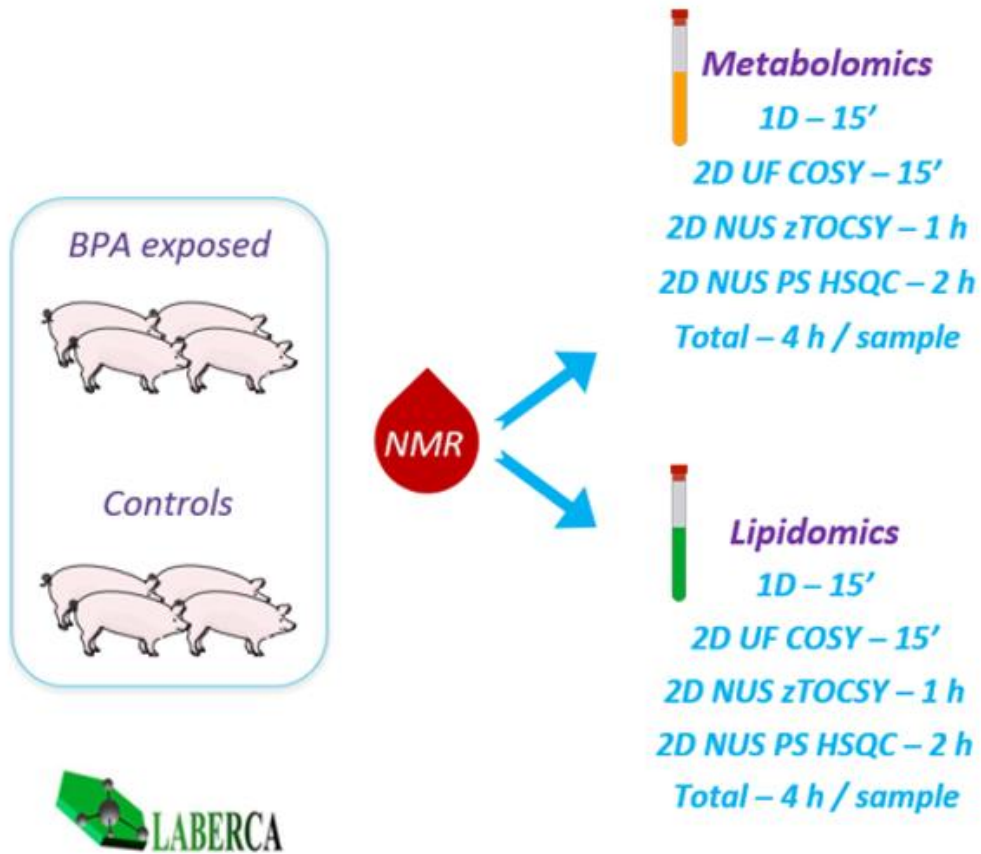
Study case



Julien Gauvreau



Annotation of the NMR spectra:

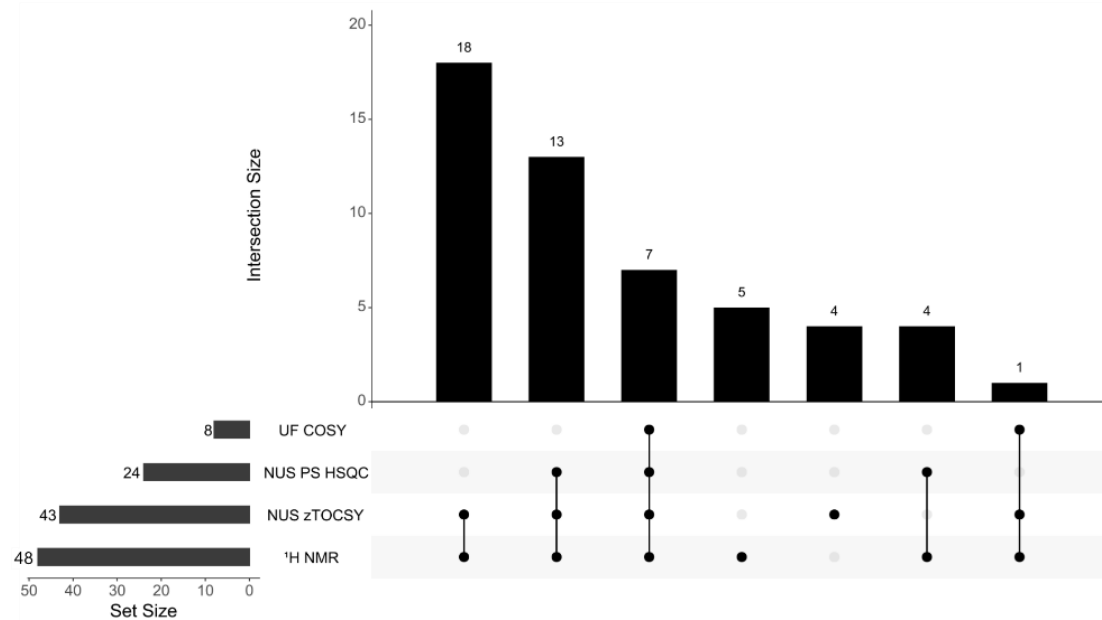


Annotation of all the signals to...

- ...develop an annotation methodology using fast 2D NMR
- ...acquire spectra on pure metabolite compounds and create a spectral database
- ...comparing the metabolite found in NMR and in MS

2D increase the confidence and the number of annotations

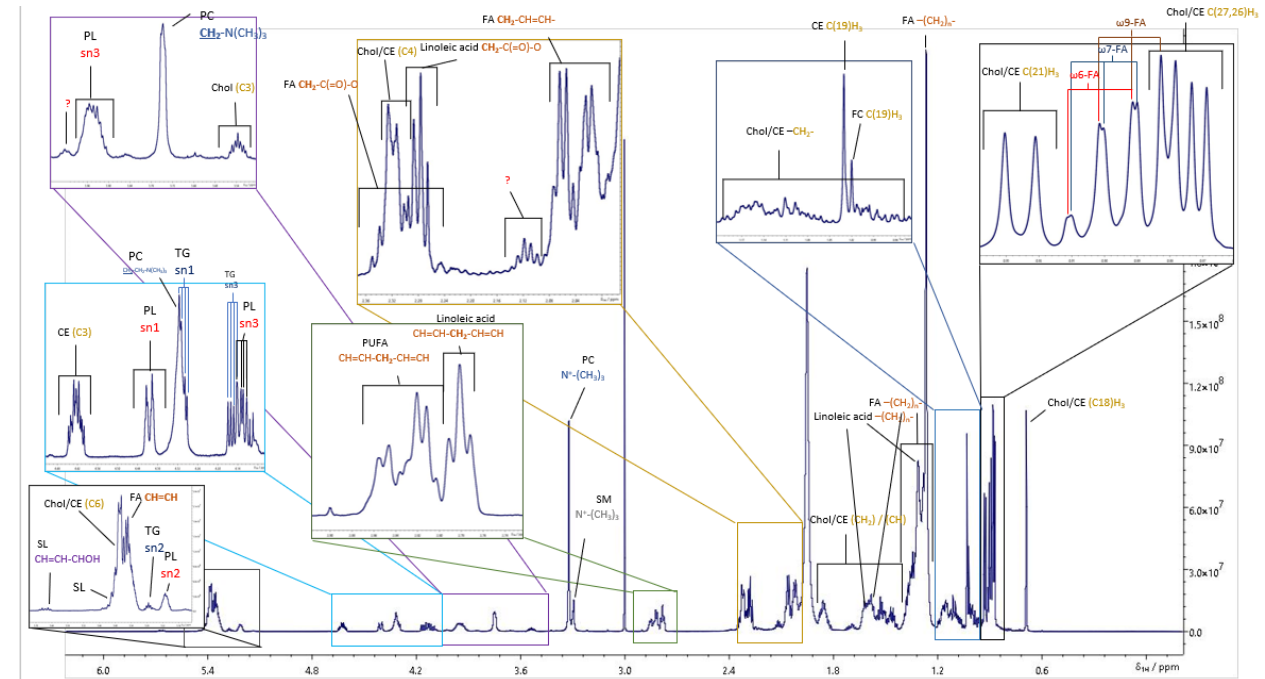
Metabolomics



Metabolite annotation intersection between NMR experiments

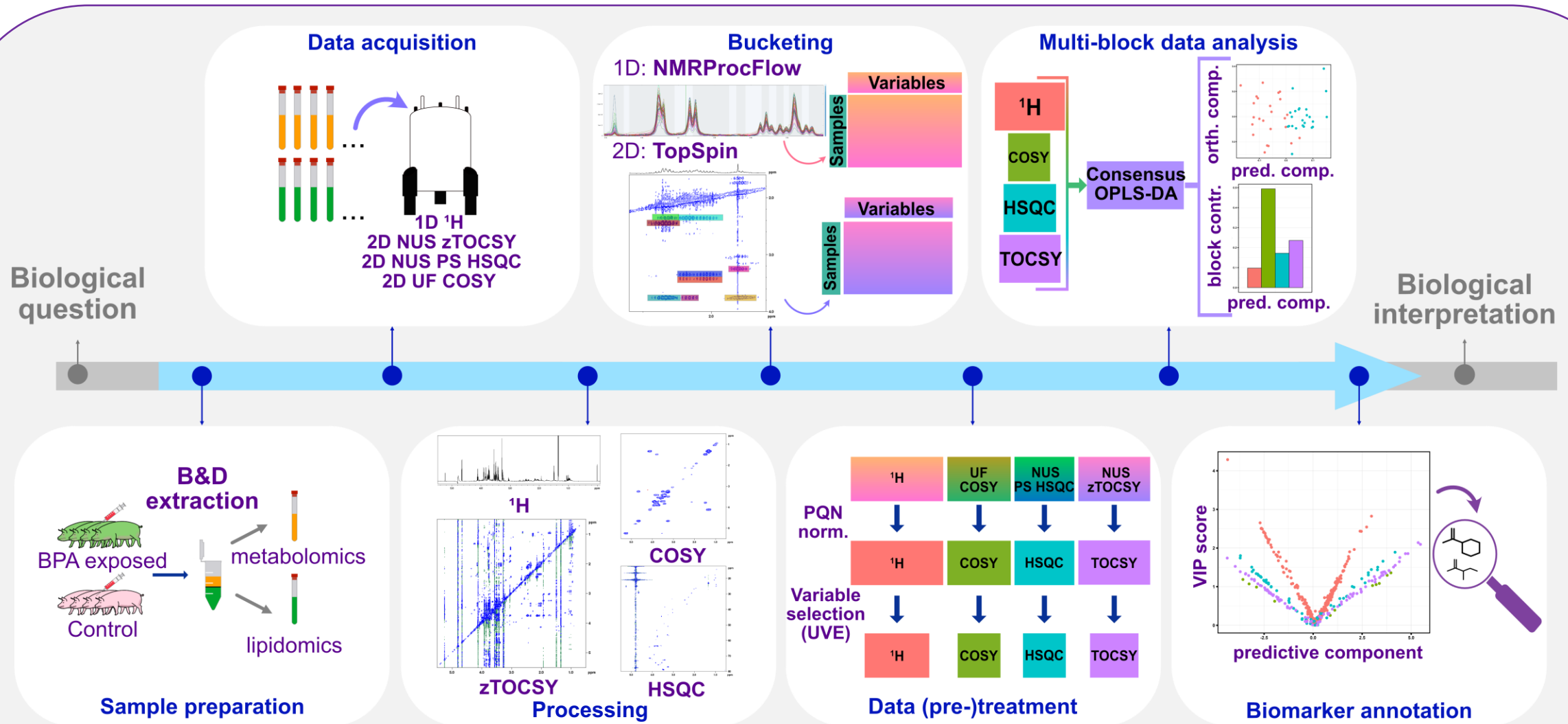
- A total of 52 compounds annotated

Lipidomics



- Annotation of 5 lipid classes :
 - Sterol
 - Phospholipids
 - Sphingolipids
 - Glycerides
 - Free Fatty acids

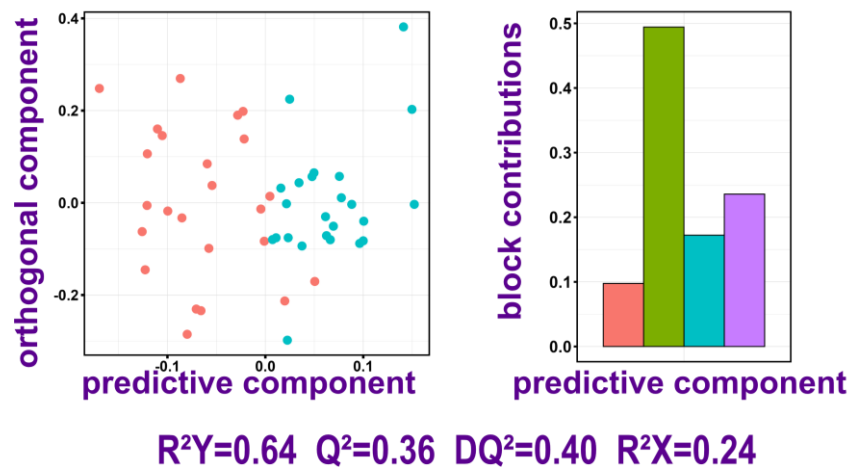
Statistical performances



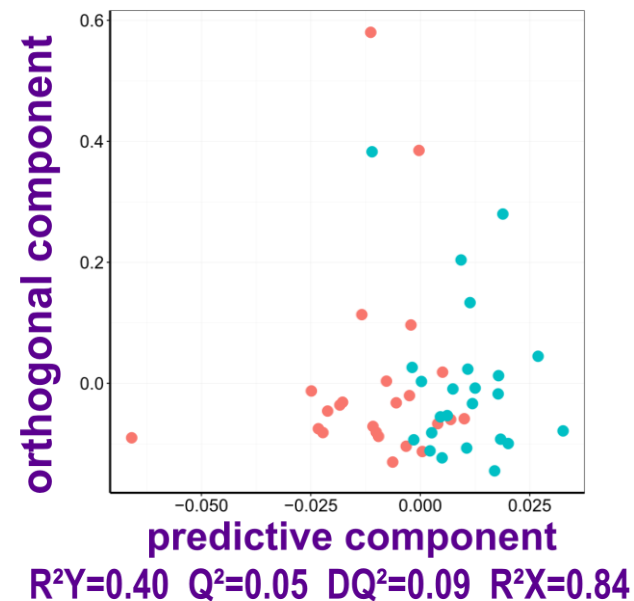
A new workflow combining both 1D and fast 2D NMR using a multi-block (MB) analysis approach

Results of the Consensus-OPLS

**Multi-block COPLS-DA
metabolomics dataset**



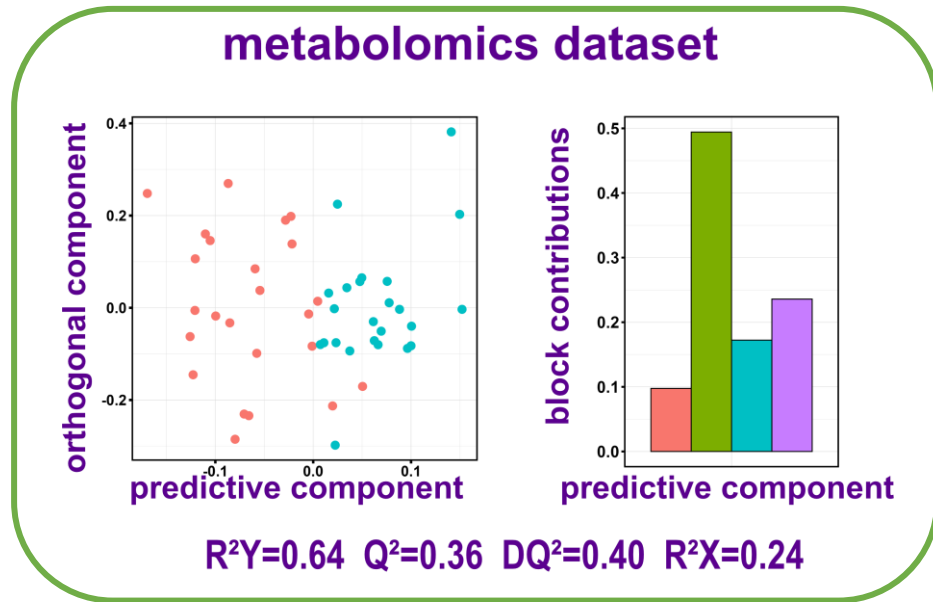
**Single-block OPLS-DA
 1H metabolomics**



Groups:
● BPA exposed
● Controls

NMR experiments:
Metabolomics:
■ 1H ■ HSQC
■ COSY ■ zTOCSY

Results of the Consensus-OPLS



Groups:

- BPA exposed
- Controls

NMR experiments:

Metabolomics:

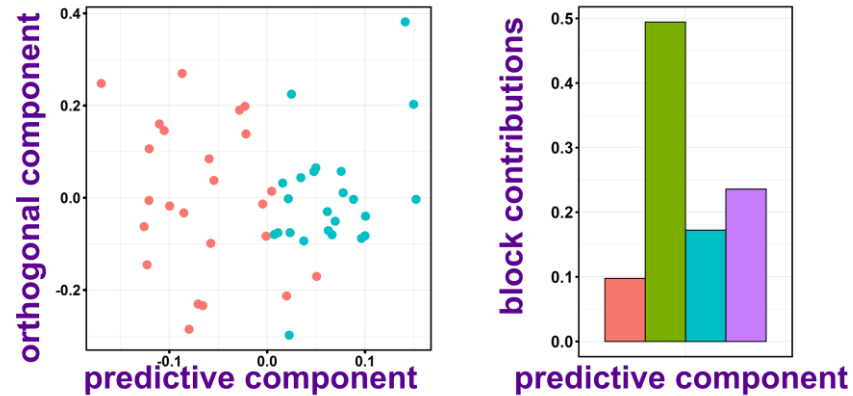
- 1H ■ HSQC
- COSY ■ zTOCSY

Lipidomics:

- 1H ■ HSQC
- COSY ■ zTOCSY

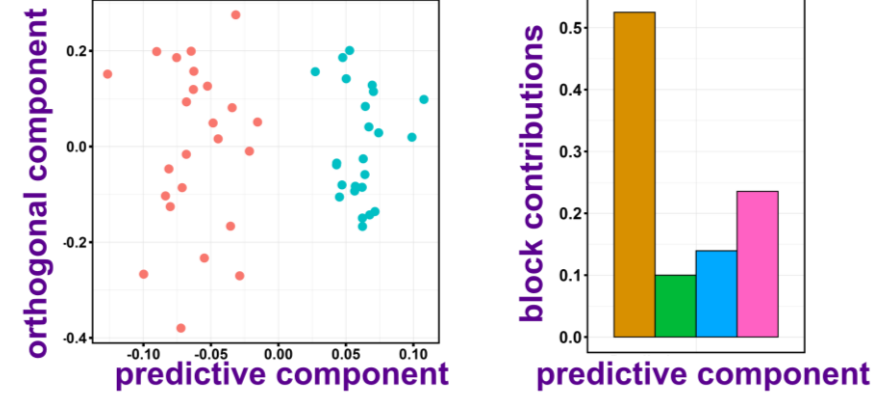
Results of the Consensus-OPLS

metabolomics dataset



$R^2Y=0.64$ $Q^2=0.36$ $DQ^2=0.40$ $R^2X=0.24$

lipidomics dataset



$R^2Y=0.89$ $Q^2=0.50$ $DQ^2=0.53$ $R^2X=0.64$

Groups:

- BPA exposed
- Controls

NMR experiments:

Metabolomics:

■ 1H ■ HSQC

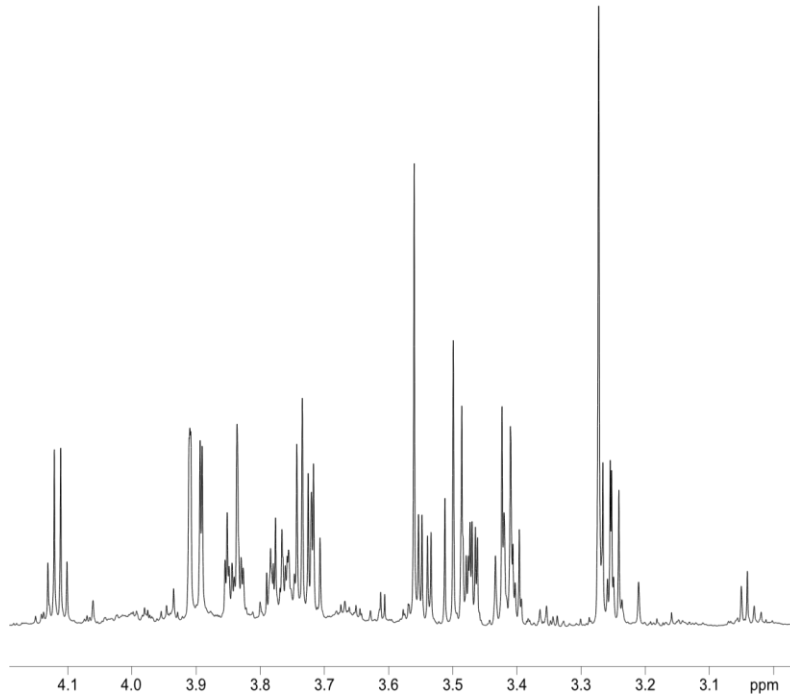
■ COSY ■ zTOCSY

Lipidomics:

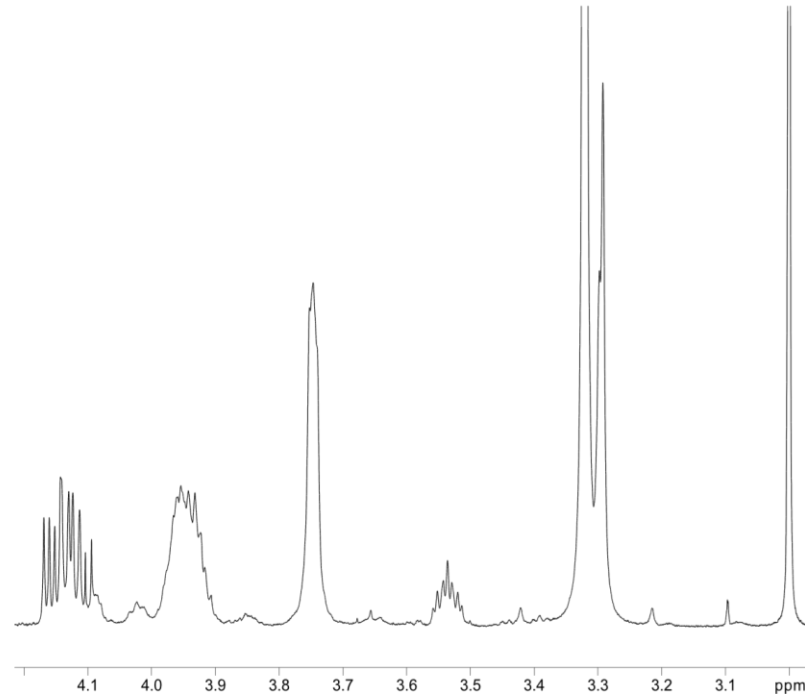
■ 1H ■ HSQC

■ COSY ■ zTOCSY

Is 2D as useful for metabolomics than lipidomics?

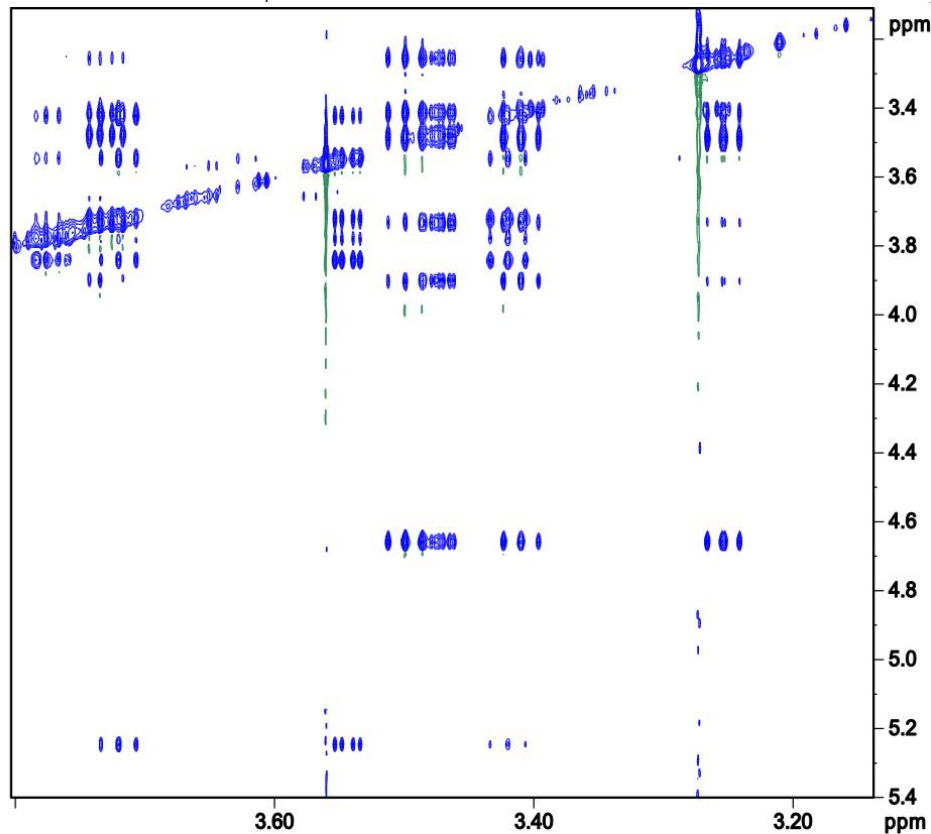
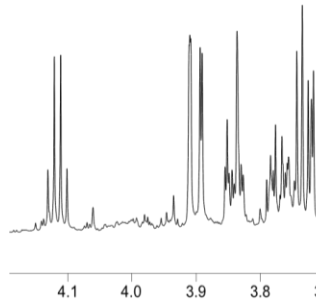


1D ^1H NMR spectrum for metabolomics

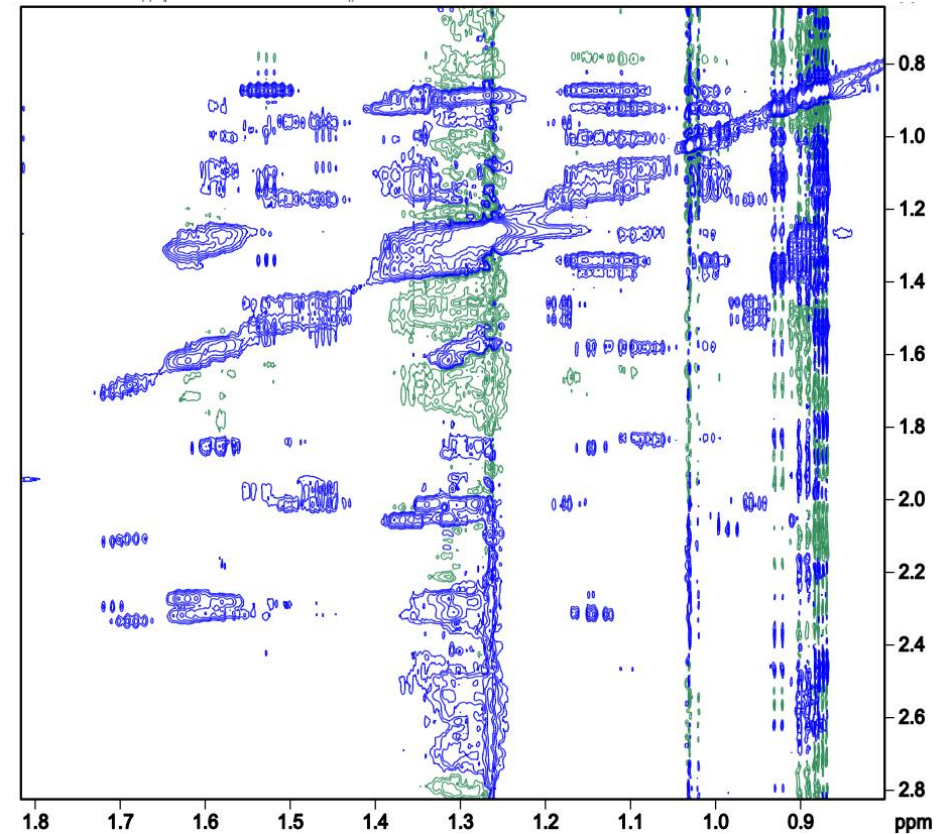


1D ^1H NMR spectrum for lipidomics

Is 2D as useful for metabolomics than lipidomics?



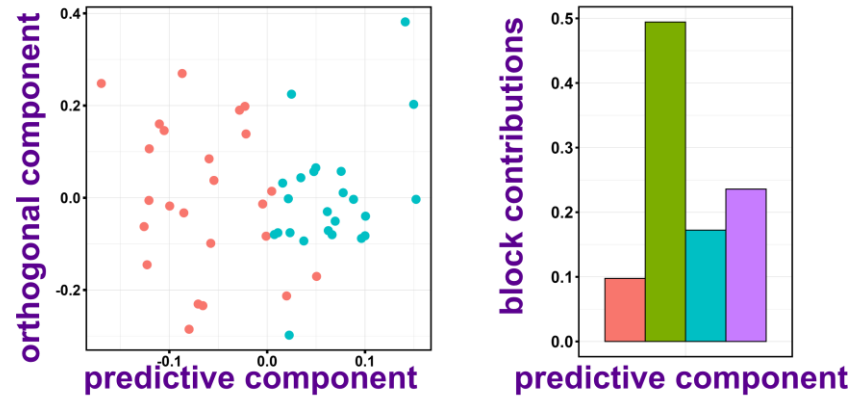
2D NUS zTOCSY spectrum for metabolomics



2D NUS zTOCSY spectrum for lipidomics

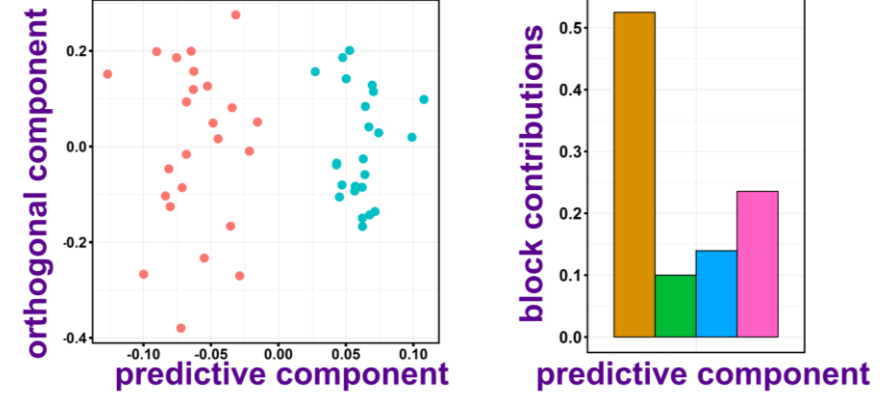
Results of the Consensus-OPLS

metabolomics dataset



$R^2Y=0.64$ $Q^2=0.36$ $DQ^2=0.40$ $R^2X=0.24$

lipidomics dataset



$R^2Y=0.89$ $Q^2=0.50$ $DQ^2=0.53$ $R^2X=0.64$

Groups:

- BPA exposed
- Controls

NMR experiments:

Metabolomics:

■ 1H ■ HSQC

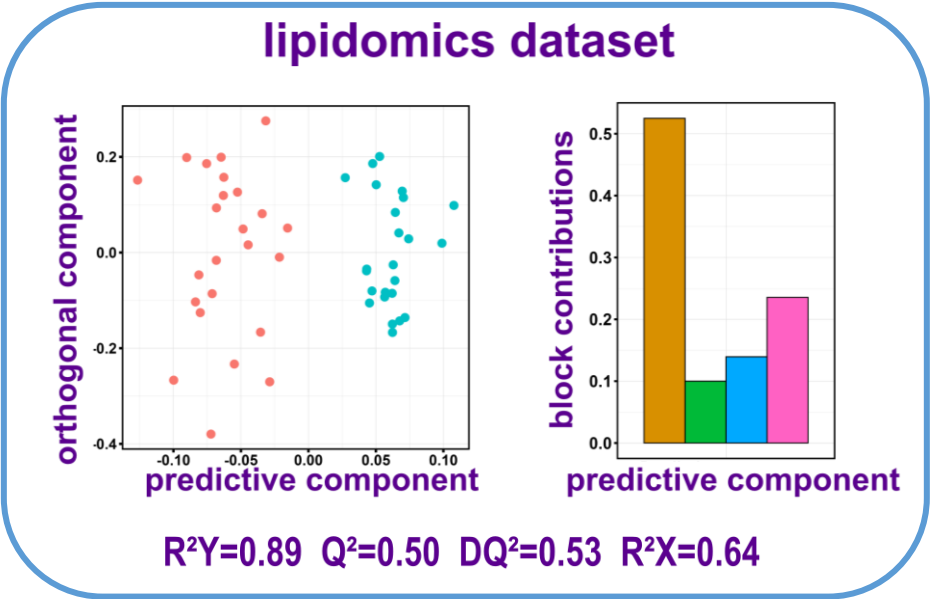
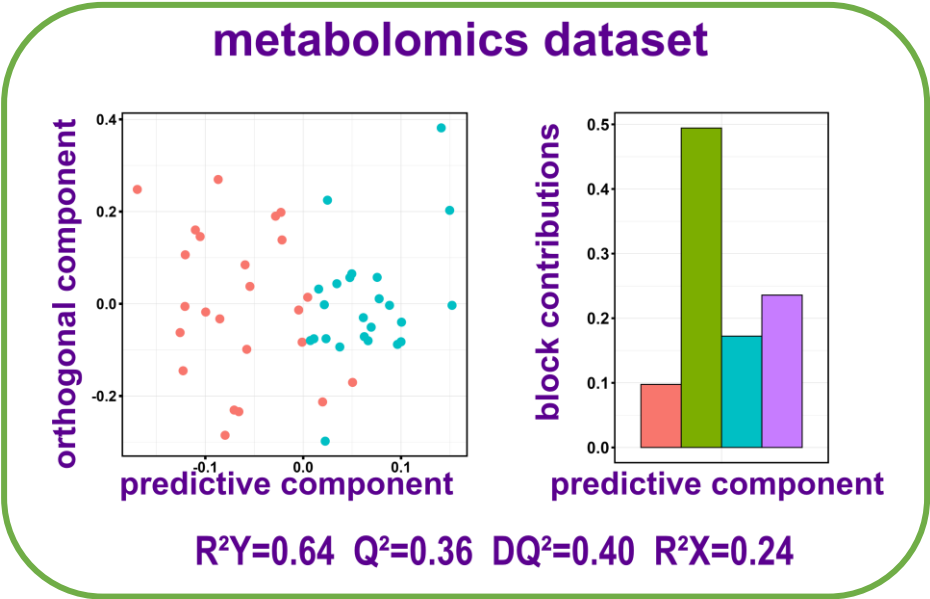
■ COSY ■ zTOCSY

Lipidomics:

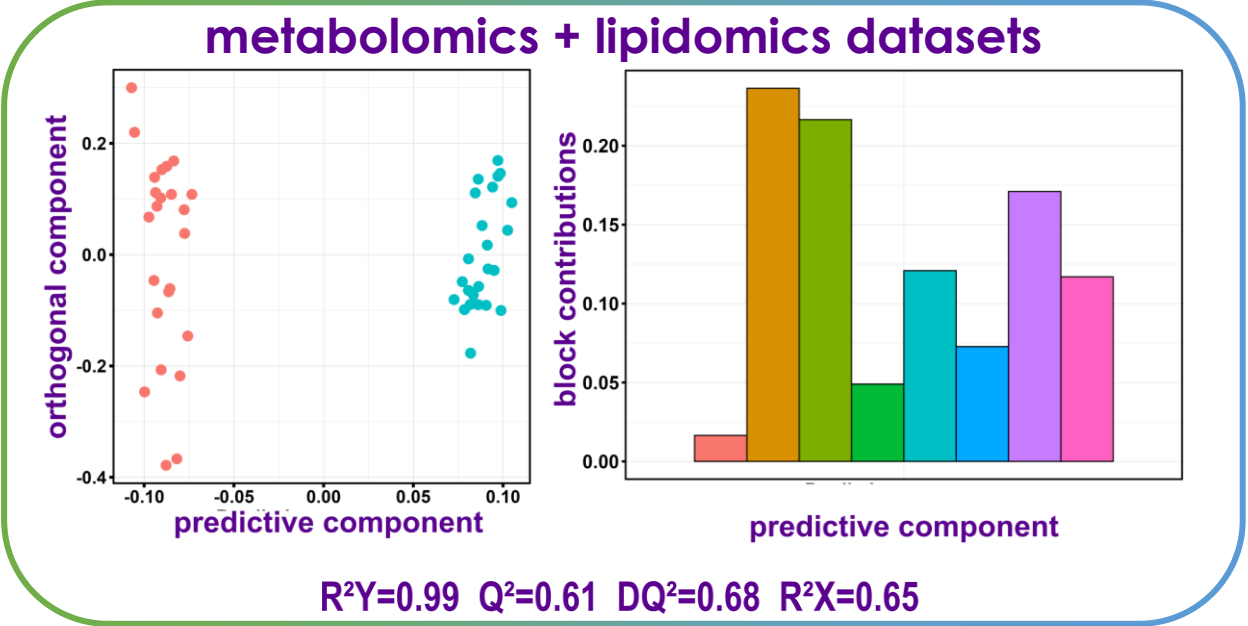
■ 1H ■ HSQC

■ COSY ■ zTOCSY

Combination of metabolomics and lipidomics

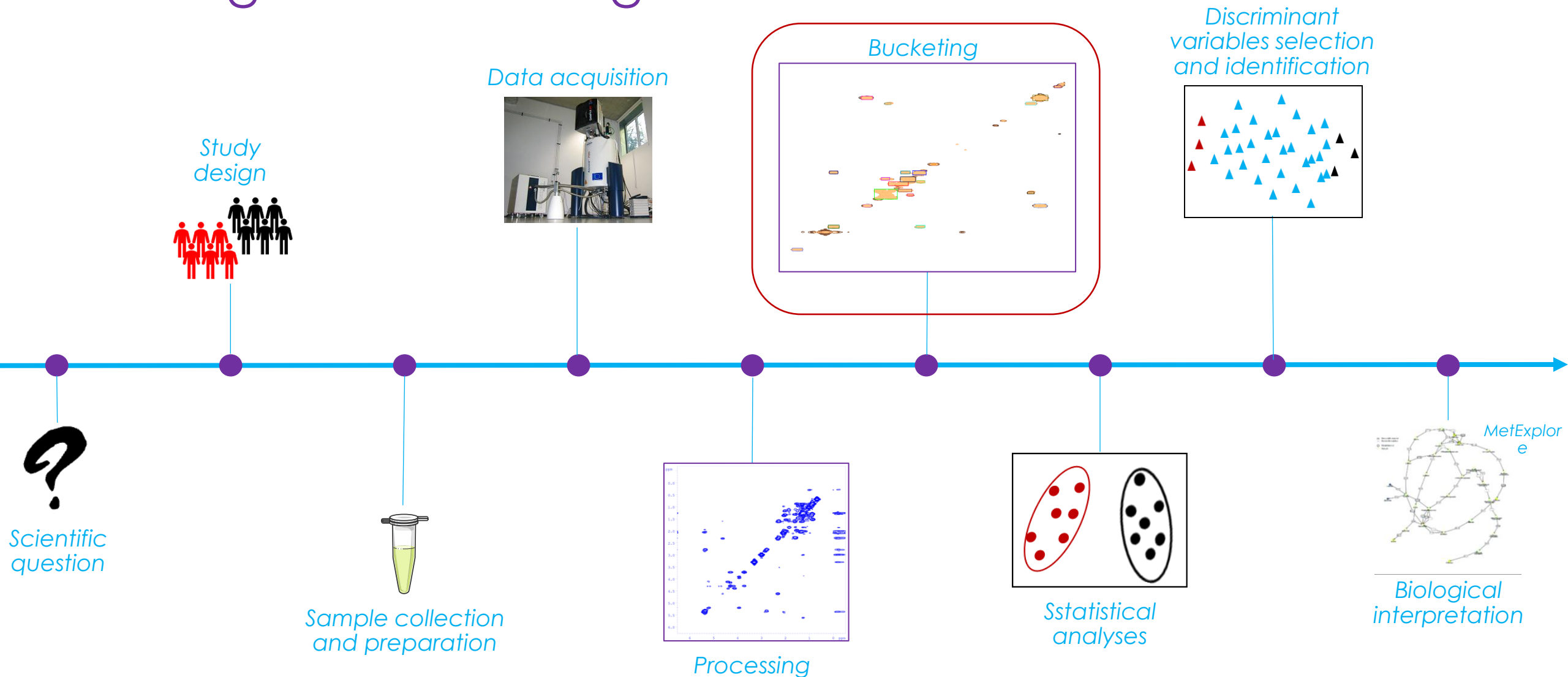


Groups:
● BPA exposed
● Controls



NMR experiments:
Metabolomics:
● 1H ● HSQC
● COSY ● zTOCSY
Lipidomics:
● 1H ● HSQC
● COSY ● zTOCSY

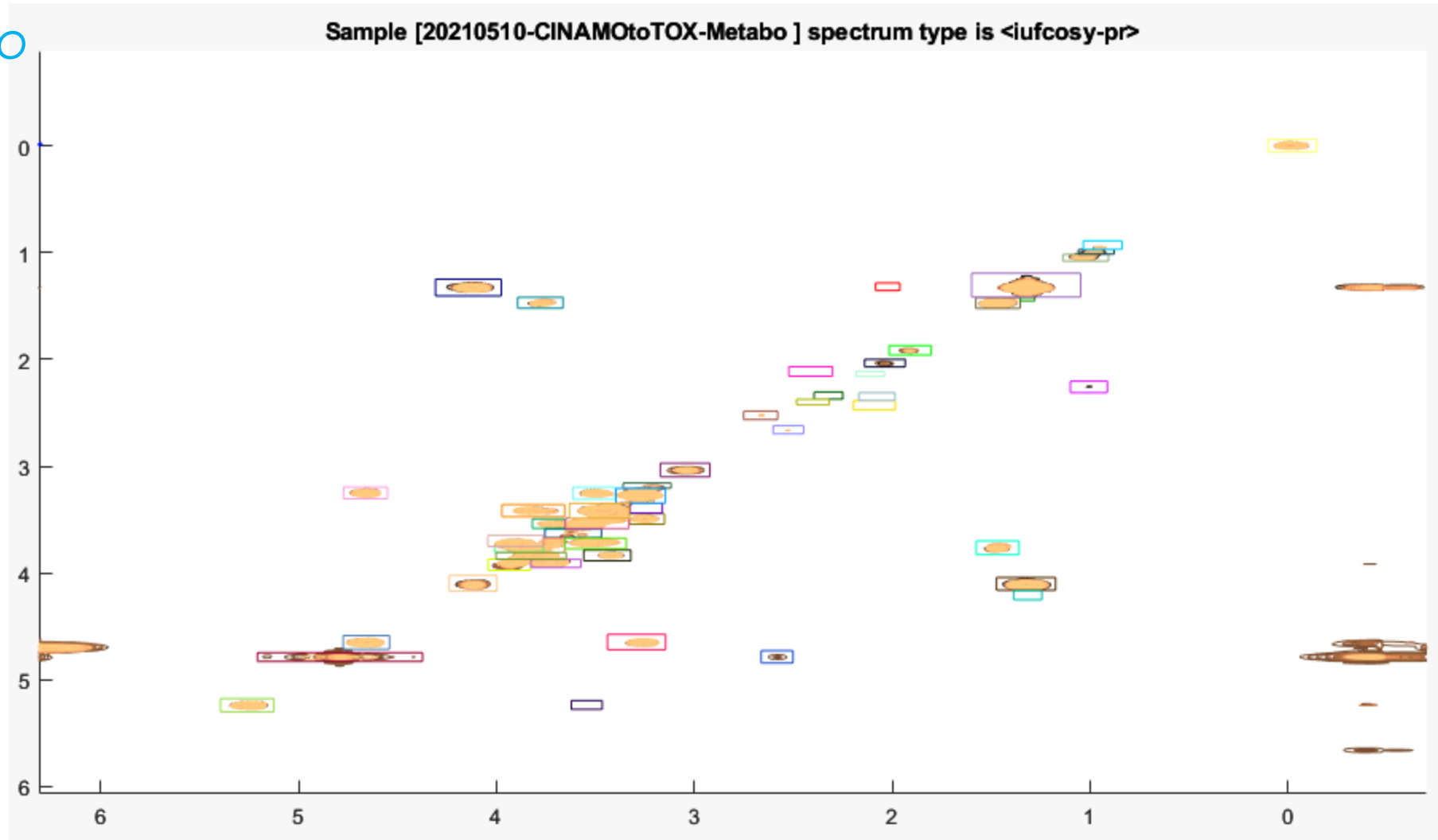
Challenge #1 remaining with 2D NMR



Integration / Bucketing / Binning of the 2D

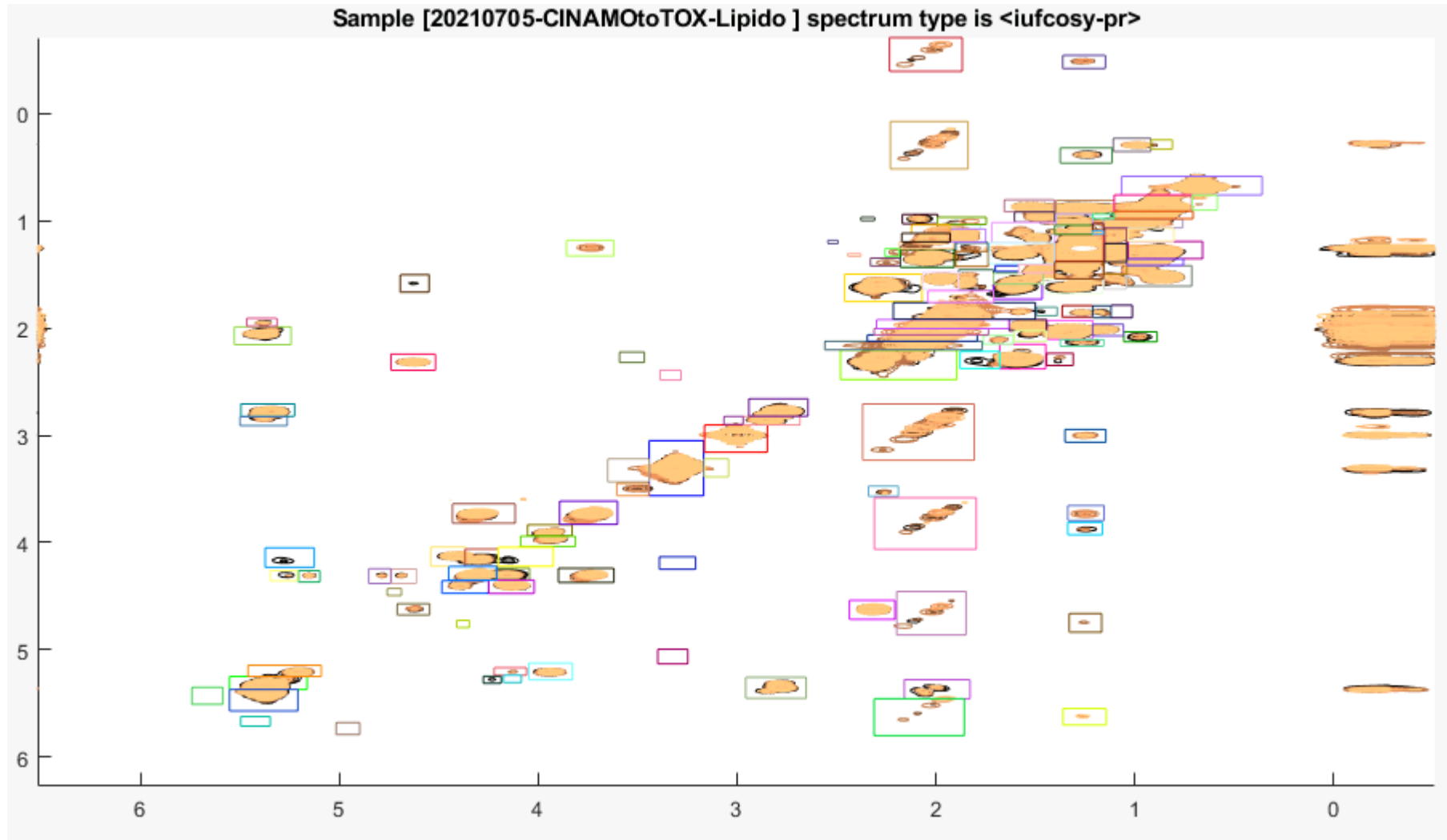
UFCOSY

Metabo



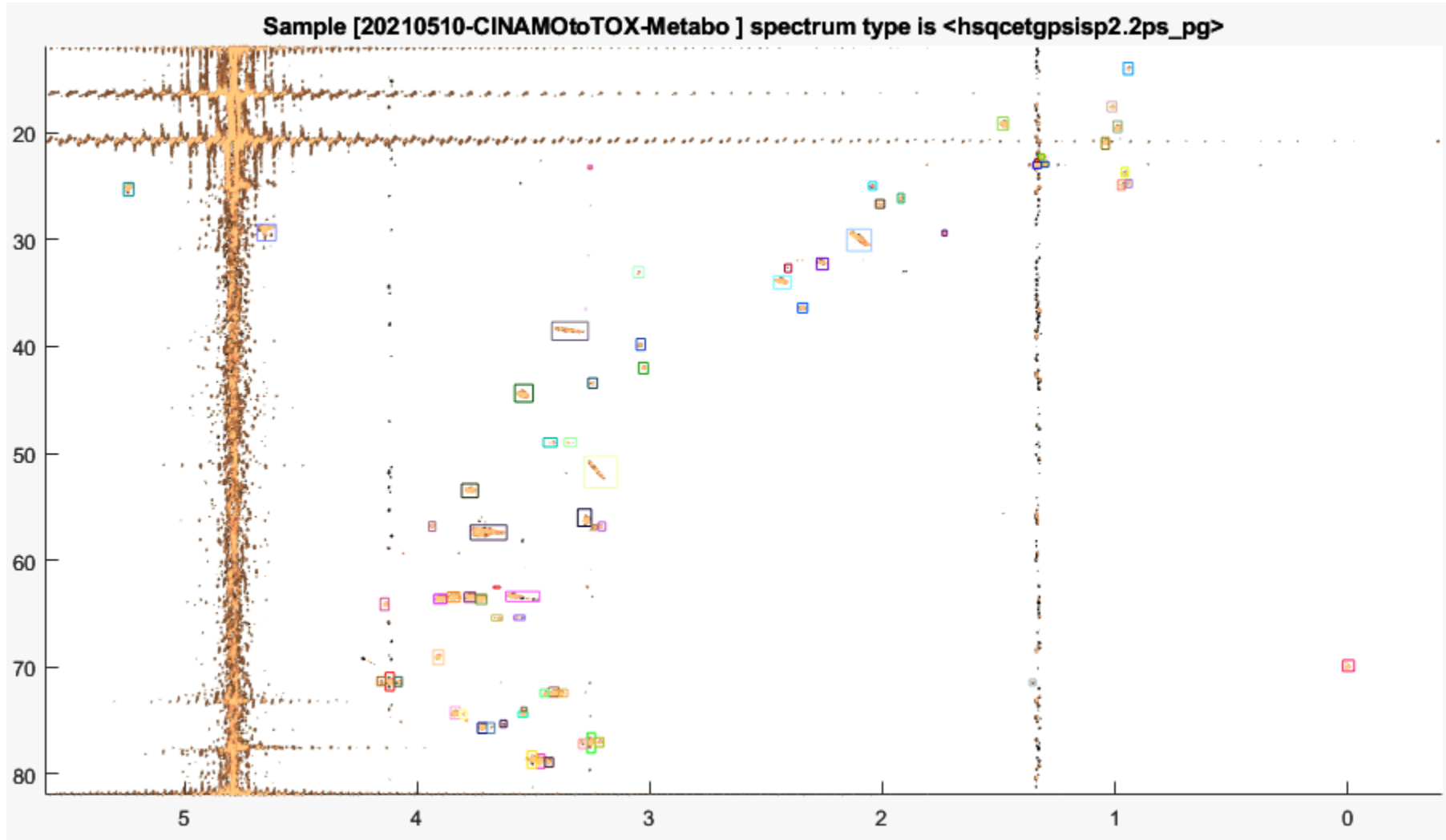
Integration / Bucketing / Binning of the 2D

UFCOSY Lipido



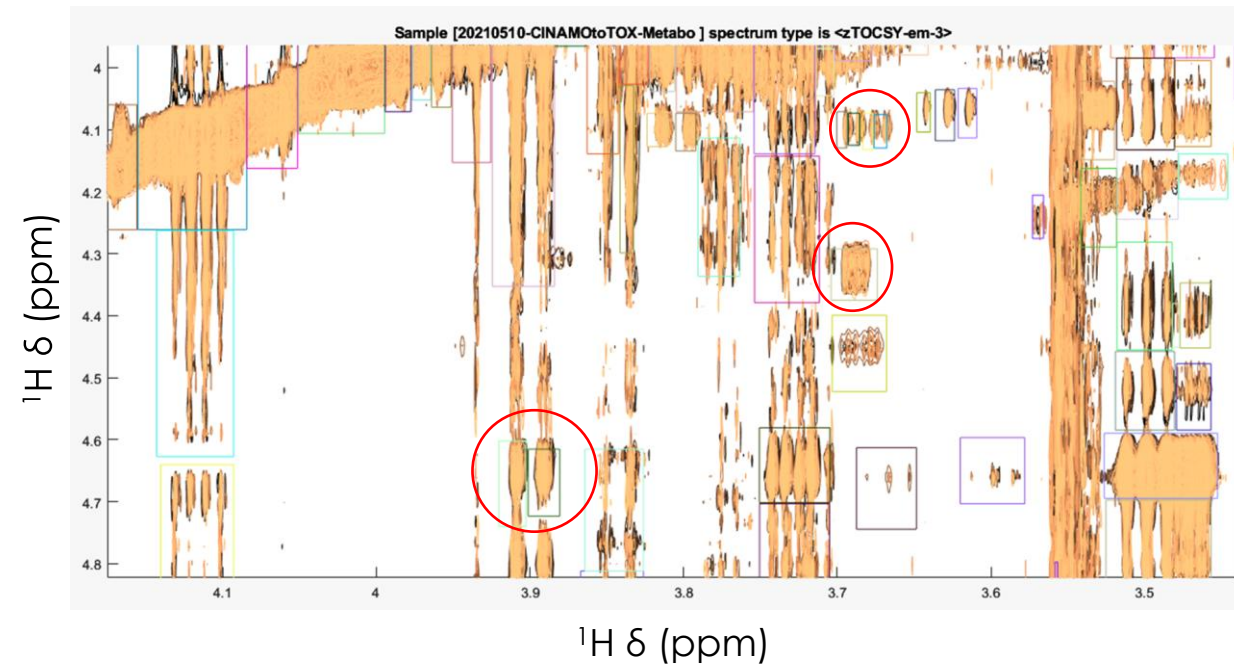
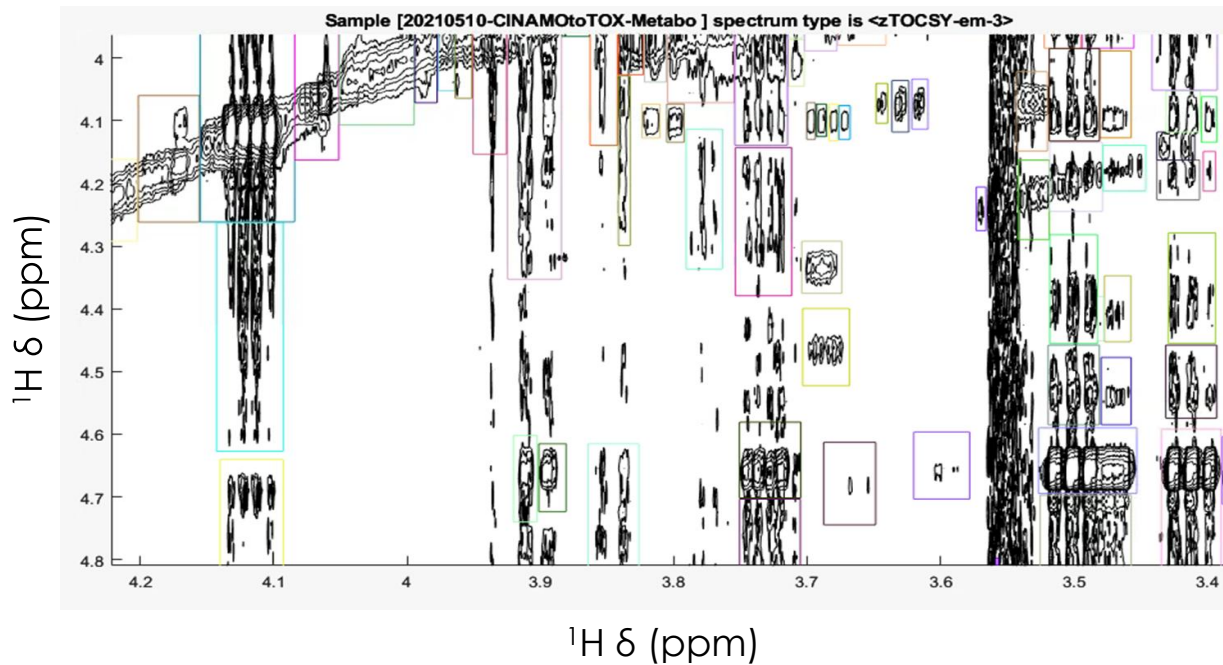
Integration / Bucketing / Binning of the 2D

PS NUS HSQC Metabo



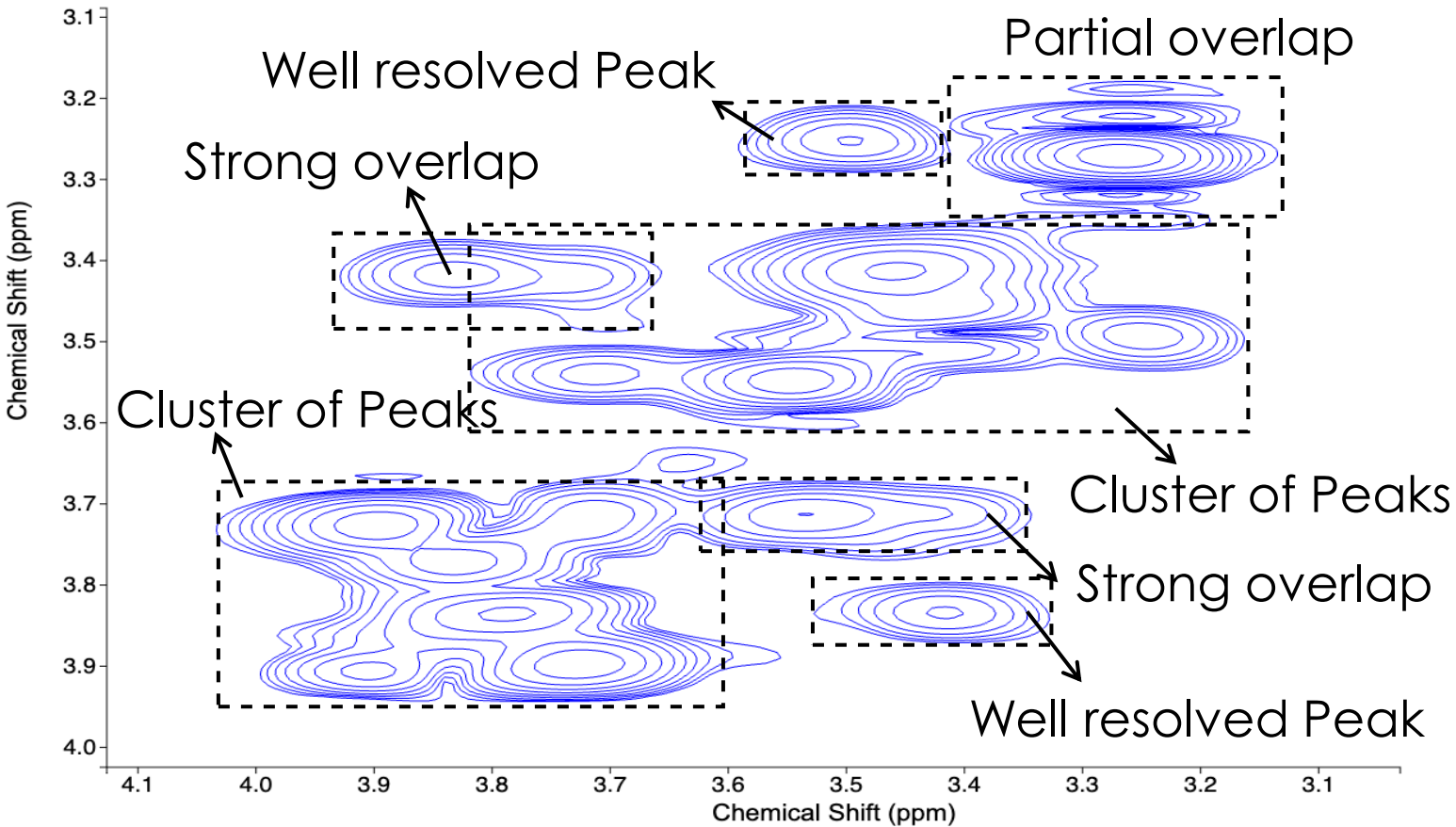
Integration / Bucketing / Binning of the 2D

NUS zTOCSY



Bucketing difficulties in batch mode when SI too high -> alignment prior bucketing?

Challenges for the integration



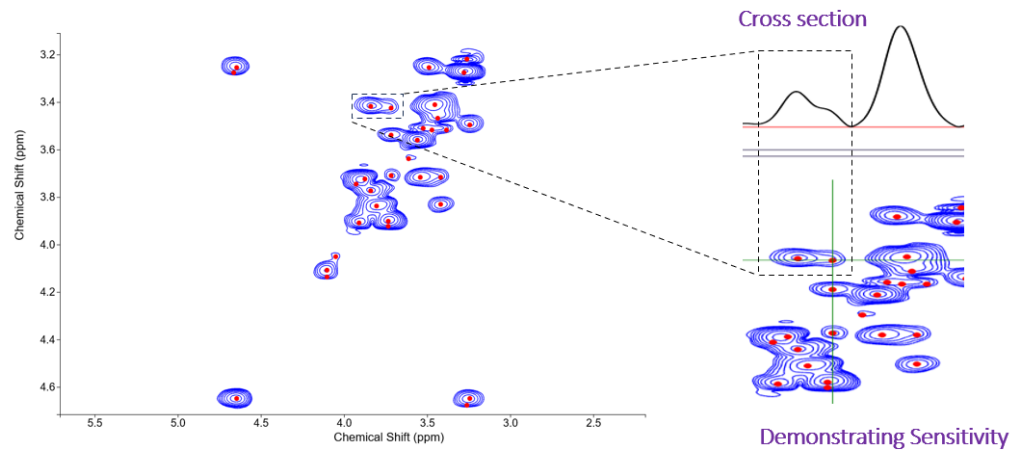
Software and Hardware challenges:

- How do we integrate many peaks
- In batch mode on dozens or hundreds of spectra
- While dealing with heavy files

2D UF COSY
zoomed-in spectral region

THE OHIO STATE
UNIVERSITY

DEEP Picker : AI-Based tool (DNN)

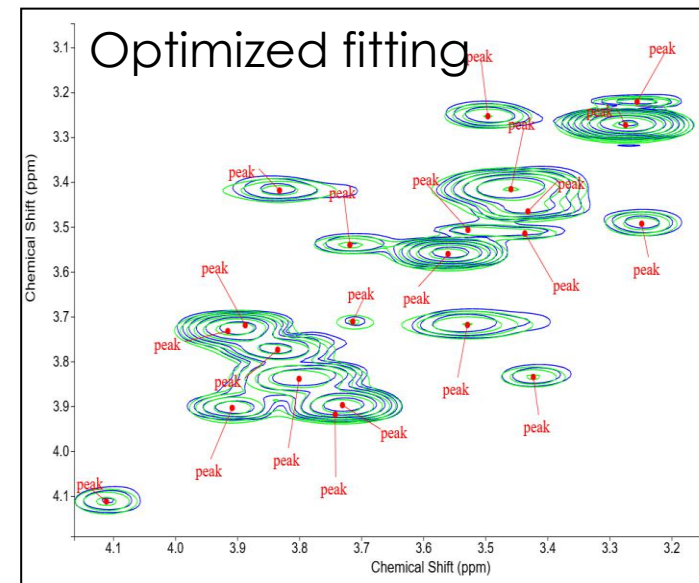
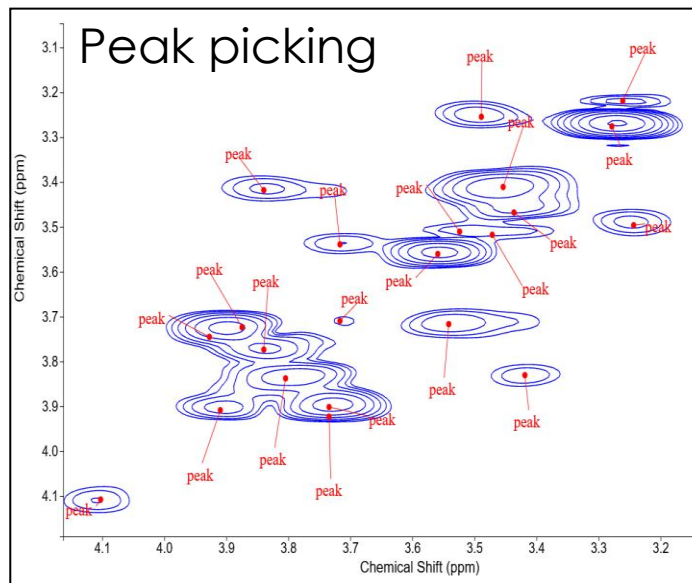


Selected region of 2D ^1H - ^1H UF COSY Spectrum

Trained Neural
Network

5000 Synthetic
Spectra

Distinguish real Peaks



Integration times



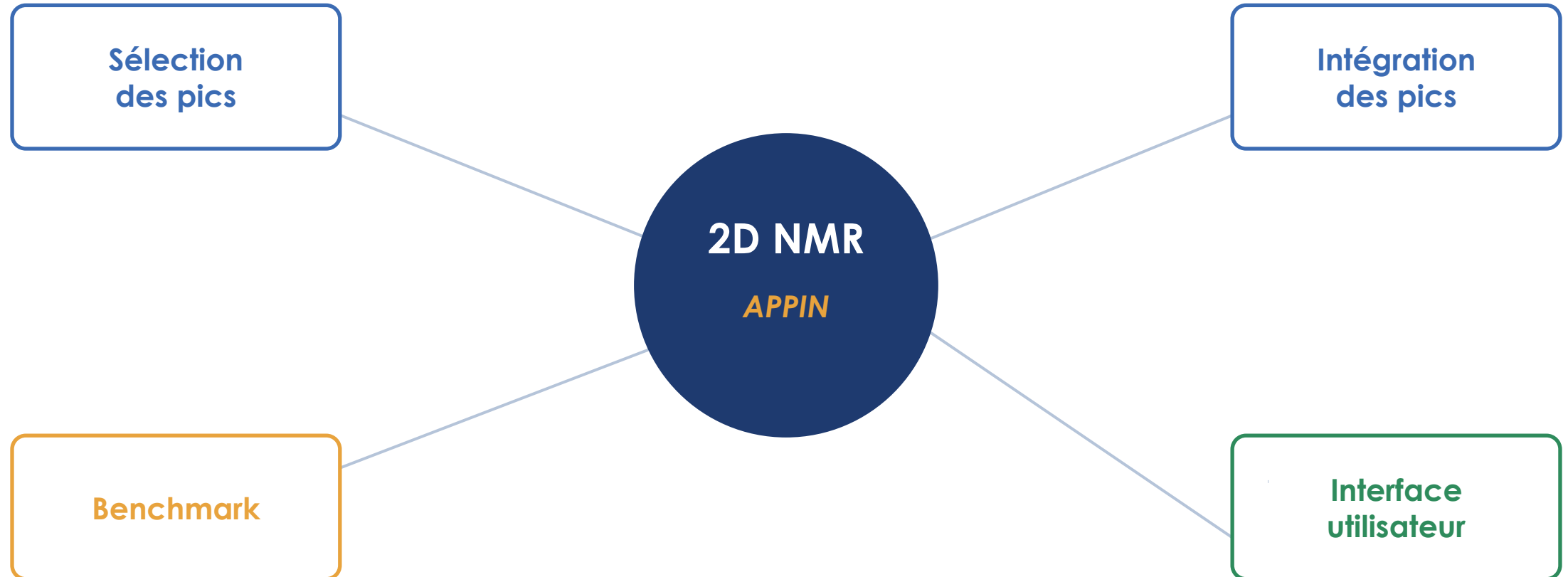
Sumit Mishra

Spectra Type/Quantity	Batch-Processing time on Ubuntu OS (Linux based)	
	Using Non-Parallel script	Using Parallel script
UF COSY/A dataset of 82-spectra	~3 hours	~30 minutes
NUS PS HSQC/A dataset of 82-spectra	~3 days	~3 hours
NUS zTOCSY/A dataset of 82-spectra	Require high configuration system	~16 hours

Optimized data matrices for even better discriminating performance

Création d'APPIN

Sélection et intégration automatiques des pics de spectres RMN 2D



Vue d'ensemble des méthodes implémentées

Sélection des pics

Maximum Local

DeepPicker / Peakviewer

CNN

Réseau convolutif (DeepPicker)

Gestion du bruit / Clustering

Seuil % max ou $\times$ niveau bruit

Intégration des pics

Sum (AUC)

Volume sous la courbe

Gaussian

NEW

Peak fitting 2D

Voigt

NEW

Pseudo-Voigt 2D

Développé en R Shiny • Interface interactive • Applicable sur des batches de 10 à 50 spectres

Sélection des pics : Maximum Local & CNN

Maximum Local

Identifie les pics en localisant les maxima d'intensité locaux dans une région définie

- **Rapide**
- Implémentation simple
- Sensible au bruit → gestion de bruit dédiée
- Référence : DeepPicker / Peakviewer

Réseau de neurones convolutif (CNN)

Analyse ligne et colonne du spectre 2D via un CNN entraîné (DeepPicker)

- **Détecte plus de pics que le maximum local**
- Robuste aux artefacts de profil
- Plus lent (passage du filtre sur tout le spectre)
- Référence : DeepPicker (Wu et al.)

Intégration des pics : 3 méthodes disponibles

Sum (AUC)

Méthode historique

Somme directe des intensités dans la zone du pic.

Avantages

- Calcul instantané
- Aucun paramètre à régler
- Robuste sur pics bien isolés

Limites

- Sensible au bruit de fond
- Pas de séparation des pics chevauchants

Gaussian

Peak fitting 2D

NEW

Ajuste chaque pic par une gaussienne 2D (algorithme nlsLM).

Avantages

- Sépare les pics chevauchants

Limites

- Plus coûteux en temps de calcul
- Convergence incertaine sur pics mal résolus

Voigt

Pseudo-Voigt 2D

NEW

Combinaison gaussienne–lorentzienne, fidèle aux pics RMN réels.

Avantages

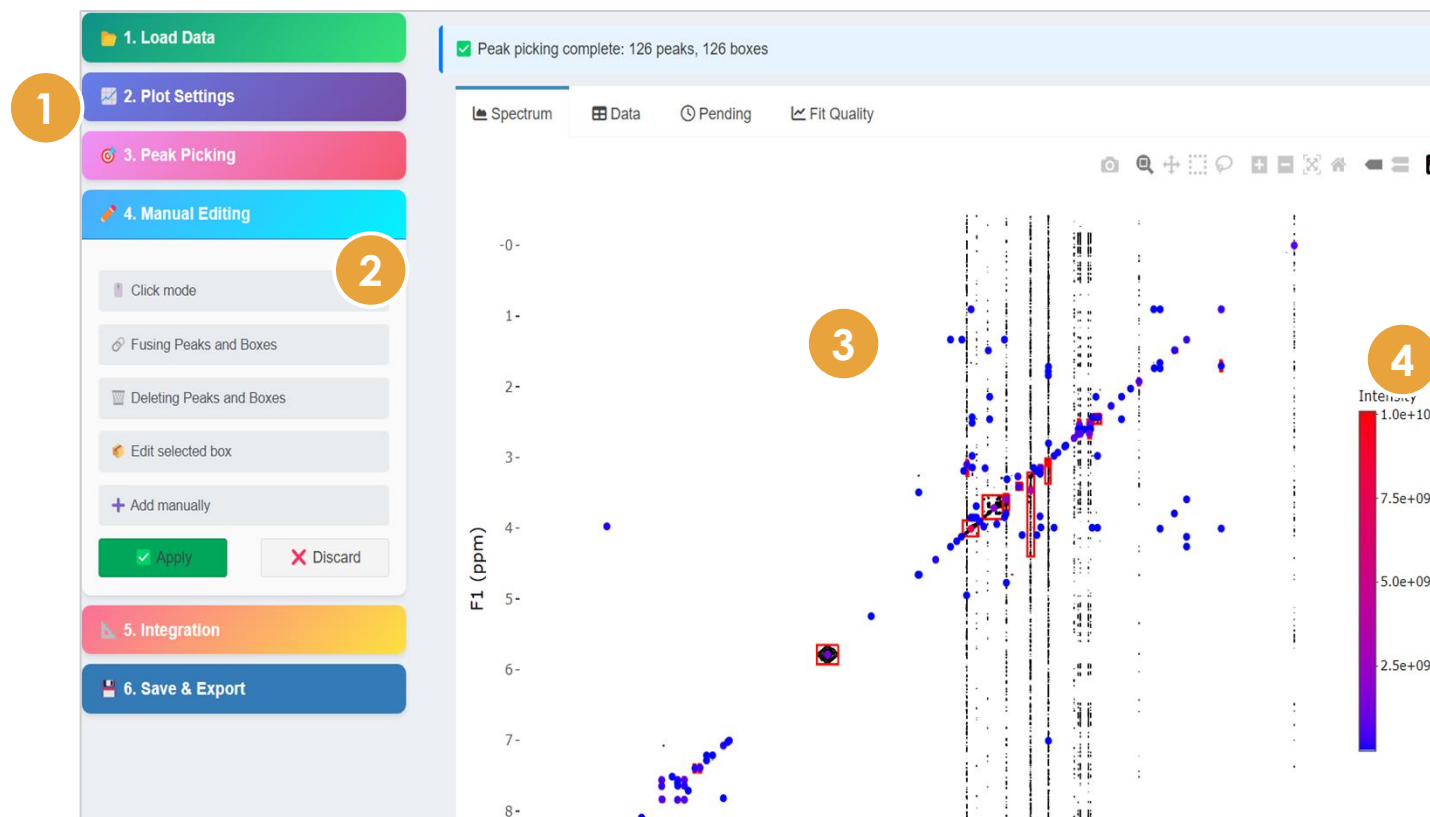
- Profil le plus réaliste en RMN

Limites

- Légèrement plus lent que le fit gaussien

→ La méthode est sélectionnable dans l'interface.

Interface APPIN : un outil interactif en R Shiny



① **Workflow en 6 étapes guidées** — chargement → peak-picking → édition → intégration → export.

② **Outils d'édition manuelle** — fusion, suppression, édition ou ajout manuel de pics et boîtes.

③ **Spectre 2D interactif** — pics détectés (points bleus) et boîtes d'intégration (rectangles rouges).

④ **Échelle d'intensité** — carte de chaleur colorée selon l'intensité des pics.

Jeu de données : ring-test MetaboHub

4

Réplicats d'urine
synthétique (SU1–SU4)

34

Métabolites quantifiés
(post-filtrage)

>10

Experts manuels
Comparaison avec 2 de ces experts
CC & JM

3

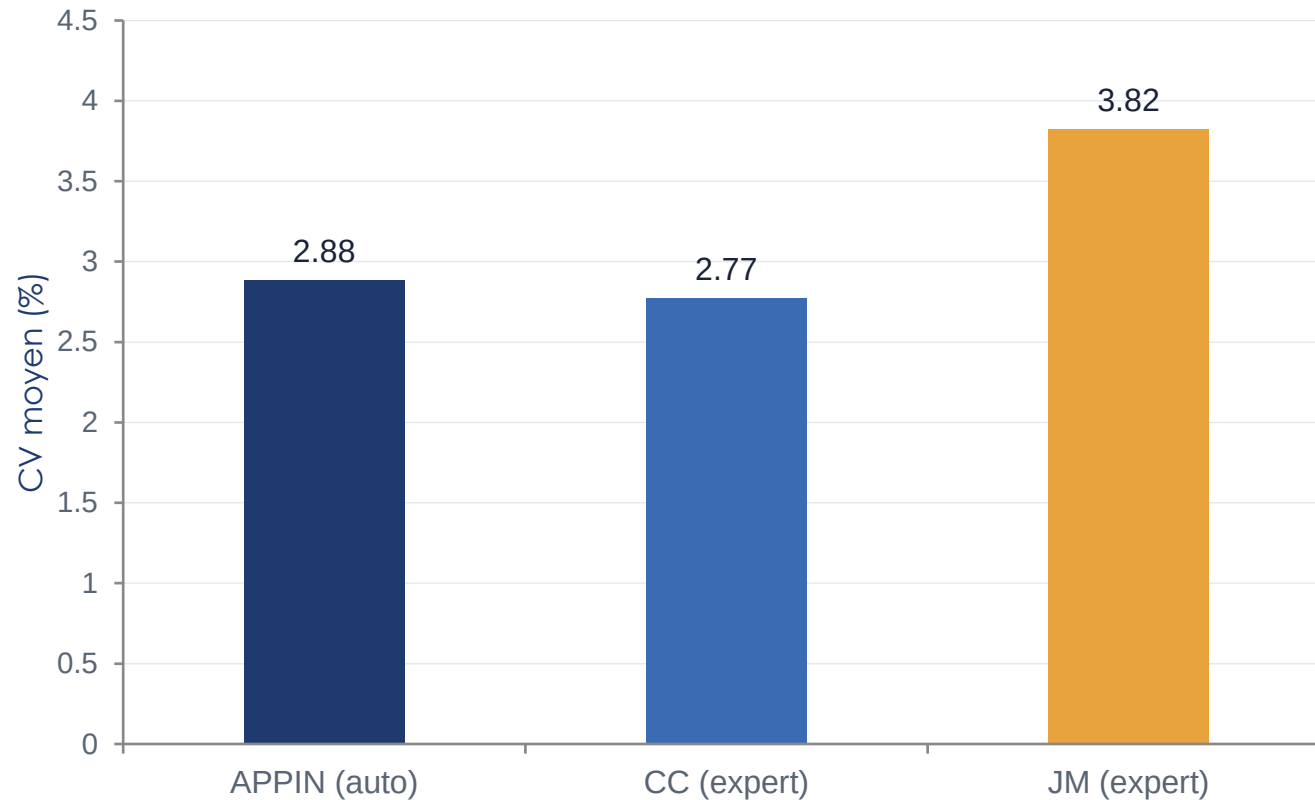
Méthodes APPIN
Sum / Gaussian / Voigt

Protocole d'évaluation

- Échantillons : 4 réplicats d'urine synthétique (SU1–SU4) — projet inter-laboratoires MetaboHub
- Acquisition : 2D ZF-TOCSY avec échantillonnage non uniforme (NUS)
- Comparaison : APPIN (intégration automatique) vs deux opérateurs experts (CC = Cécile Canlet, JM = Jérémie Marchand)
- Quantification : calibration externe (5 standards C1–C5) — régression linéaire par métabolite, déduction des concentrations
- Métriques : CV intra-méthode (précision) et concordance APPIN ↔ experts (justesse relative)

Résultat 1 : Précision (CV intra-méthode)

Coefficient de variation moyen sur les 4 réplicats SU1–SU4 — métabolite par métabolite, puis moyenne



À retenir

APPIN = 2.88 %

Au niveau du meilleur opérateur expert (CC)

- Précision comparable aux experts
- Sur les 34 métabolites présents

→ APPIN atteint une précision équivalente à un opérateur expert sans intervention manuelle

Résultat 2 : Concordance APPIN vs experts

Corrélation des concentrations estimées sur les 34 métabolites communs

APPIN vs CC

$r = 0,9986$

Coefficient de corrélation de Pearson

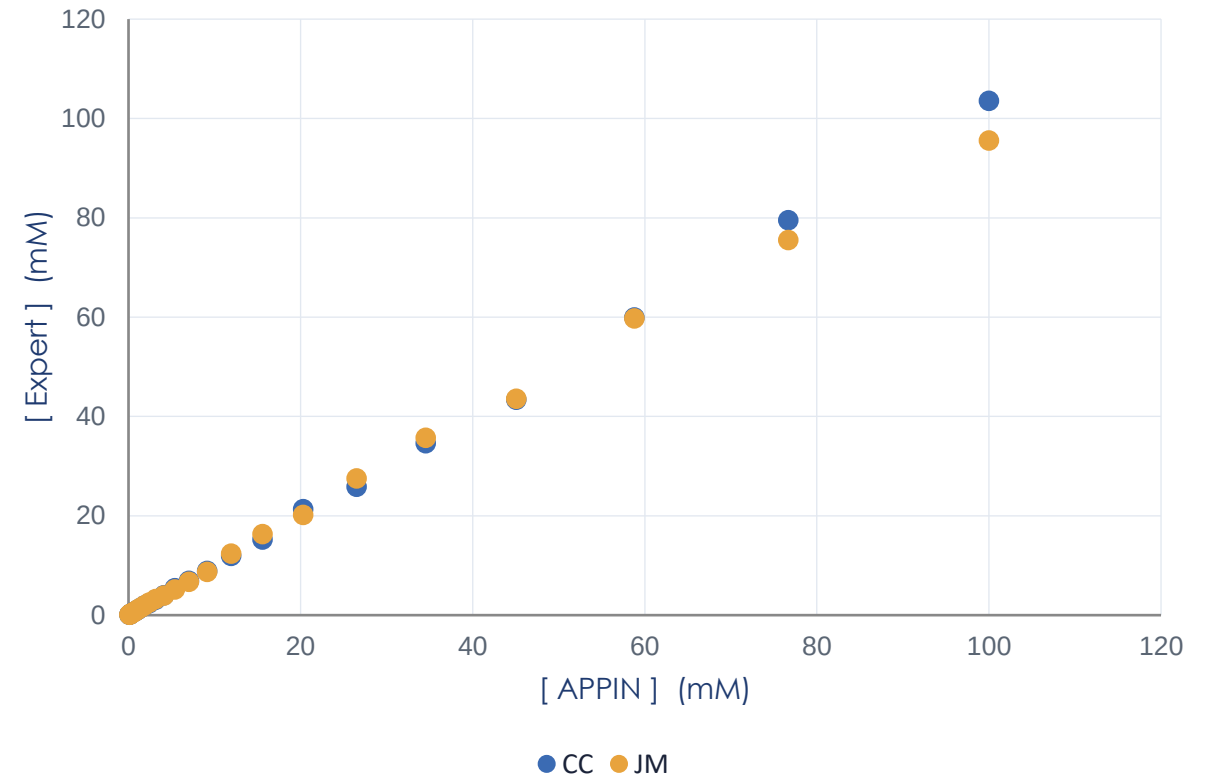
→ 28 / 34 métabolites à $\pm 10\%$

APPIN vs JM

$r = 0,9987$

Coefficient de corrélation de Pearson

→ 28 / 34 métabolites à $\pm 10\%$



→ Très forte concordance avec les deux opérateurs experts ($r > 0,998$)

Conclusion & Perspectives

✓ Acquis

- Pipeline complet : sélection → multiplets → intégration
- 3 méthodes d'intégration : Sum, Gaussian, Voigt (peak fitting 2D)
- Précision équivalente à un opérateur expert (CV $\approx$ 2,9 %)
- Très forte concordance avec experts ($r > 0,998$)
- Interface R Shiny opérationnelle
- Outil disponible sur la forge INRAe

→ Perspectives

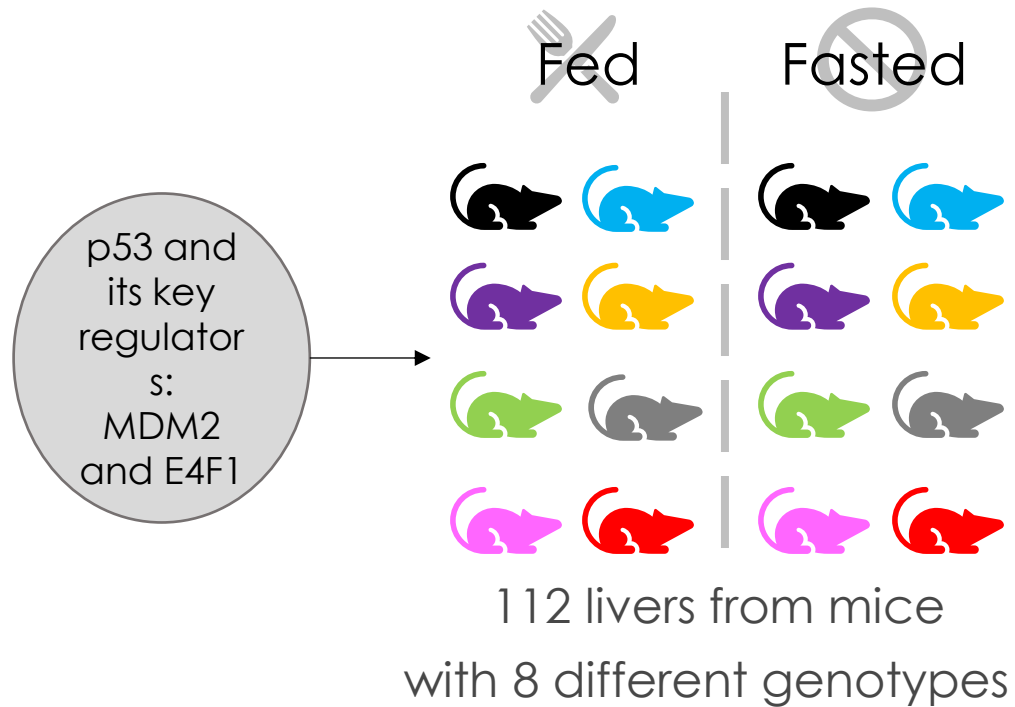
- Validation sur échantillons reels en cours — use-case avec des données de foie
- Optimisation continue de la méthode basée sur les reseaux de neurone
- **Publication méthodologique : Software paper & Use case**

APPIN est désormais prêt pour une utilisation routinière en métabolomique RMN 2D

Use-case : projet fil rouge santé de MetaboHUB



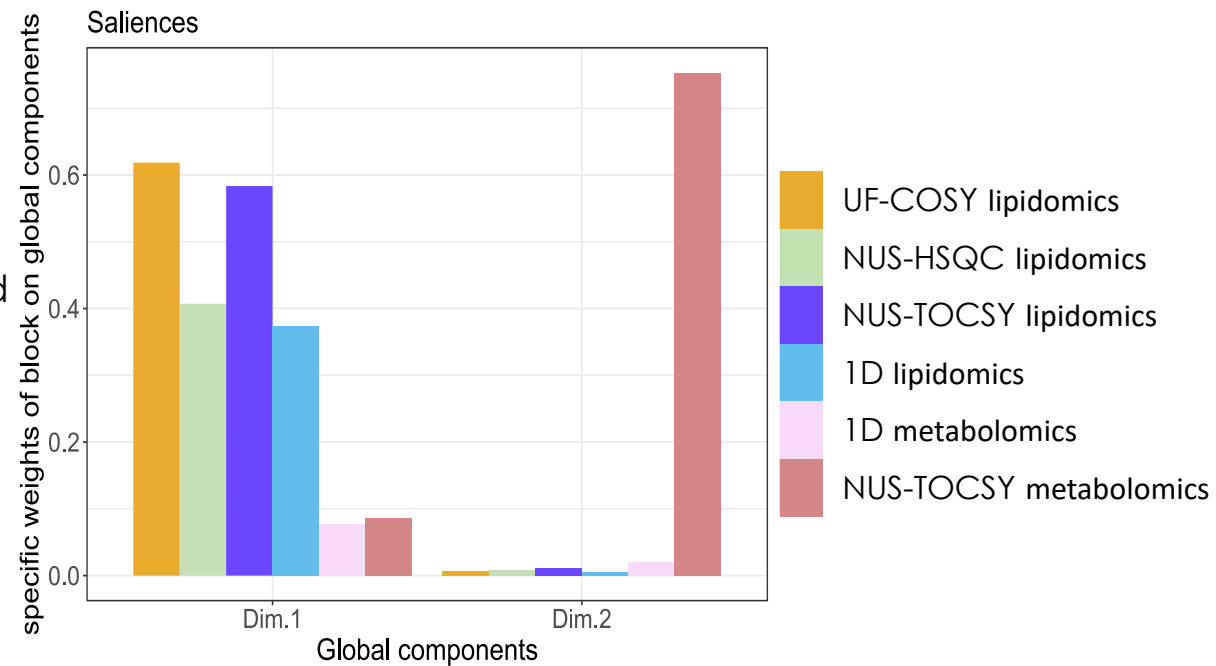
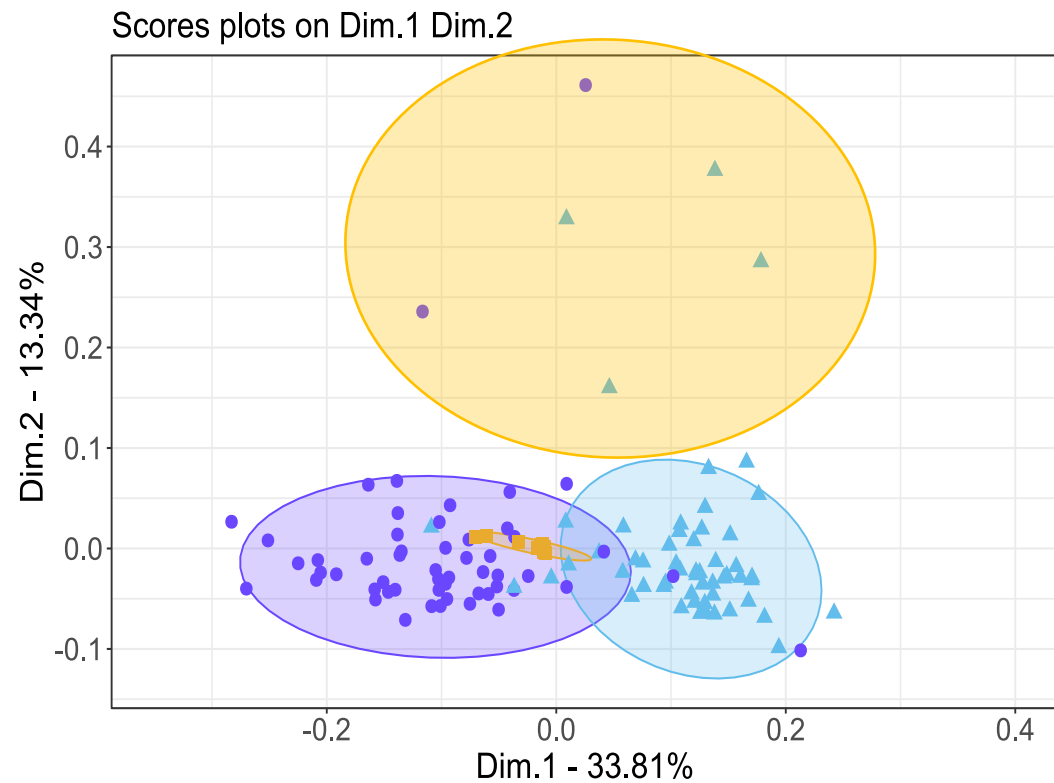
Marie Delage



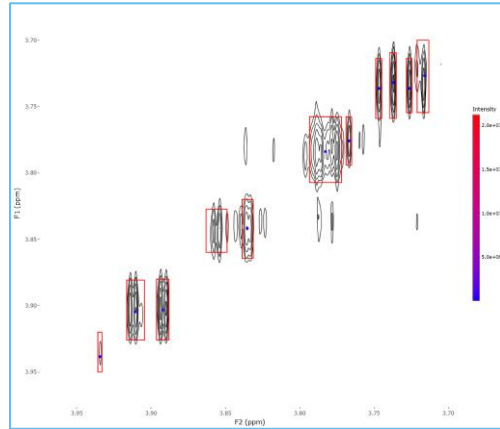
1. Wild Type (WT)
2. Single Knock Out (SKO) of **E4F1**
3. SKO of **MDM2**
4. SKO of **P53**
5. Double Knock Out (DKO) of **E4F1** and **P53**
6. DKO of **E4F1** and **MDM2**
7. DKO of **MDM2** and **P53**
8. Triple Knock Out (TKO)

APPIN successfully applied...
ALTHOUGH!

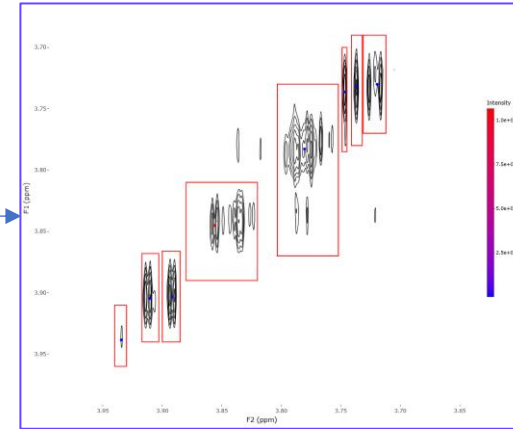
Still some problems with our NUS TOCSY



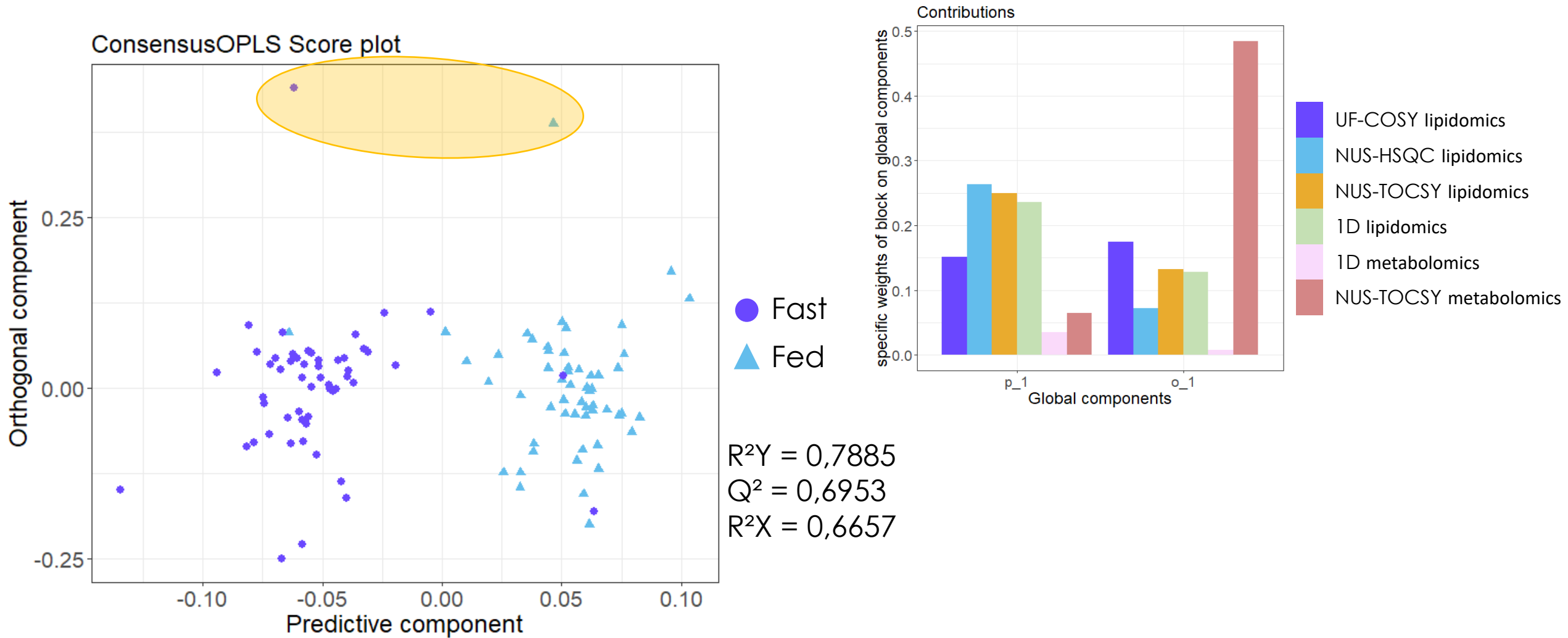
New buckets



Correction on the 6 outliers

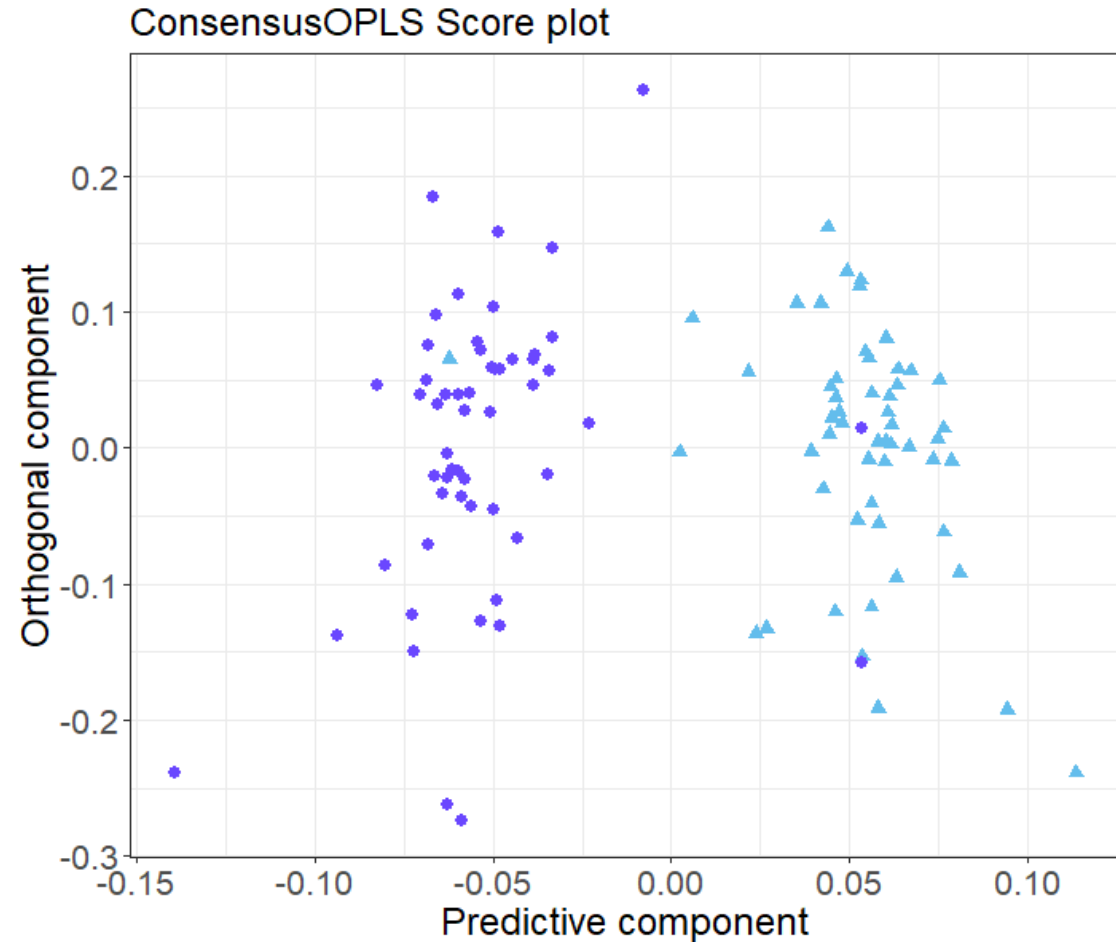


Statistical performances very sensitive to the bucketing

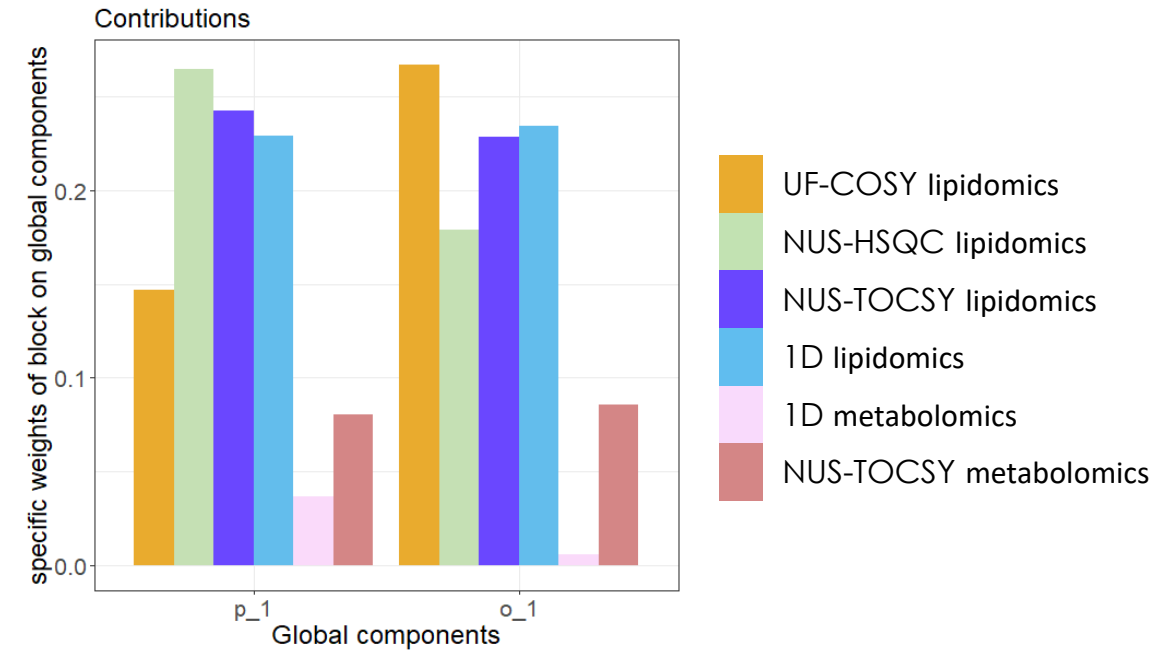


Buckets corrigées pour outliers

Results without the outliers...



$R^2Y = 0,802$
 $Q^2 = 0,6897$
 $R^2X = 0,7402$

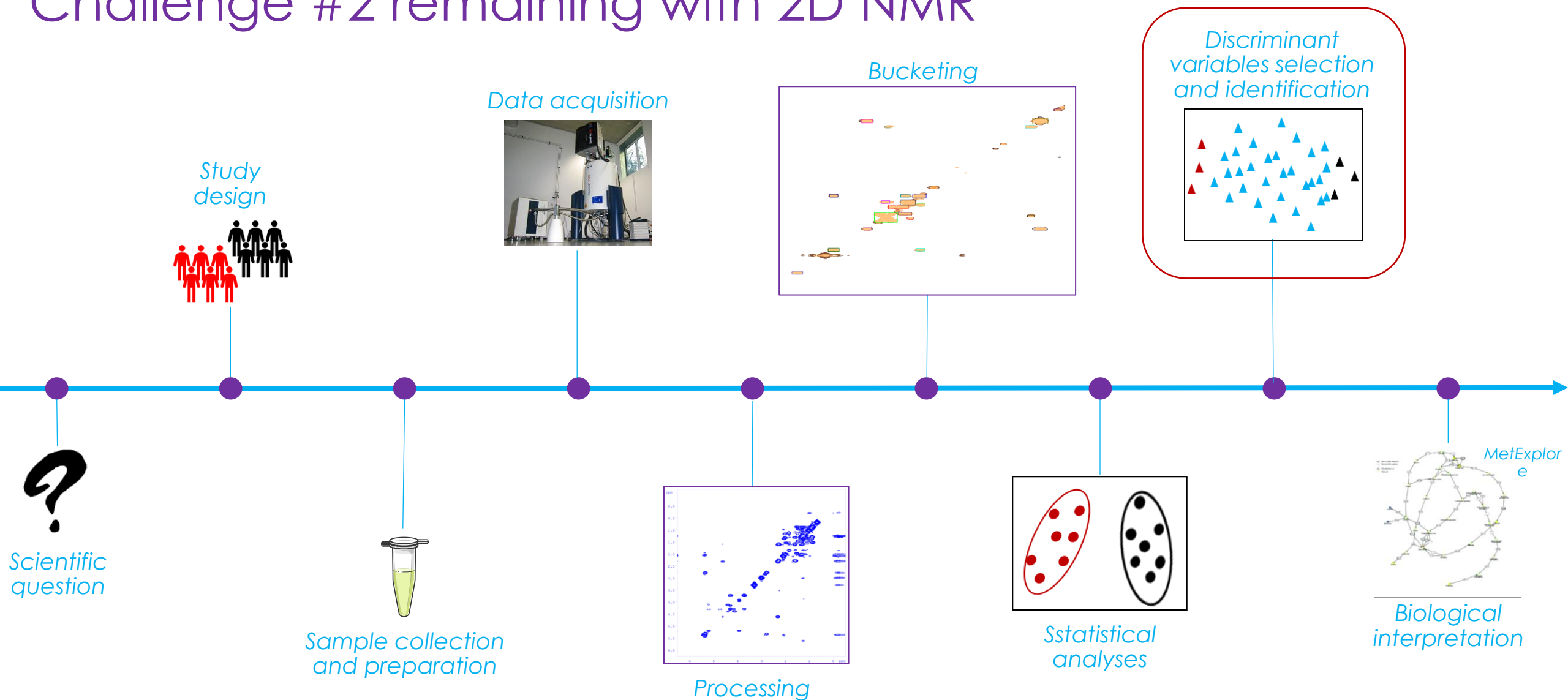


Conclusion on fast 2D NMR:

- Definitely helps statistical performances for metabolomics
- Is it really worth it for lipidomics?

Buckets corrigées pour outliers

Challenge #2 remaining with 2D NMR



Serigne SECK



Serigne Seck

NMRSCOUT

Automated Annotation of Complex Mixture by multi-dimensional NMR Using Predicted Multidimensional Data and Statistical Filtering

PROJECT BACKGROUND AND OBJECTIVES

Key challenges



No robust annotation pipeline

Need for a reproducible workflow for multidimensional NMR data.



Limited experimental 2D spectra

Restricted reference databases for COSY and HSQC spectra.



Manual annotation is not feasible

Large sample series require automated annotation strategies.

Objectives

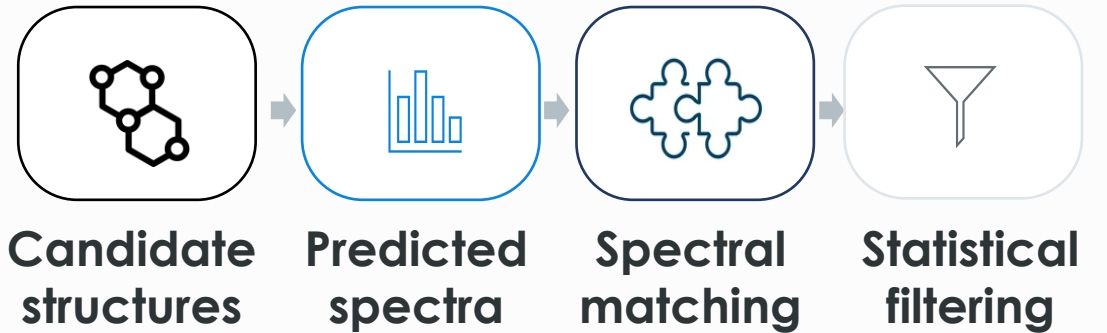
Developed tool: **NMRSCOUT**

NMR prediction

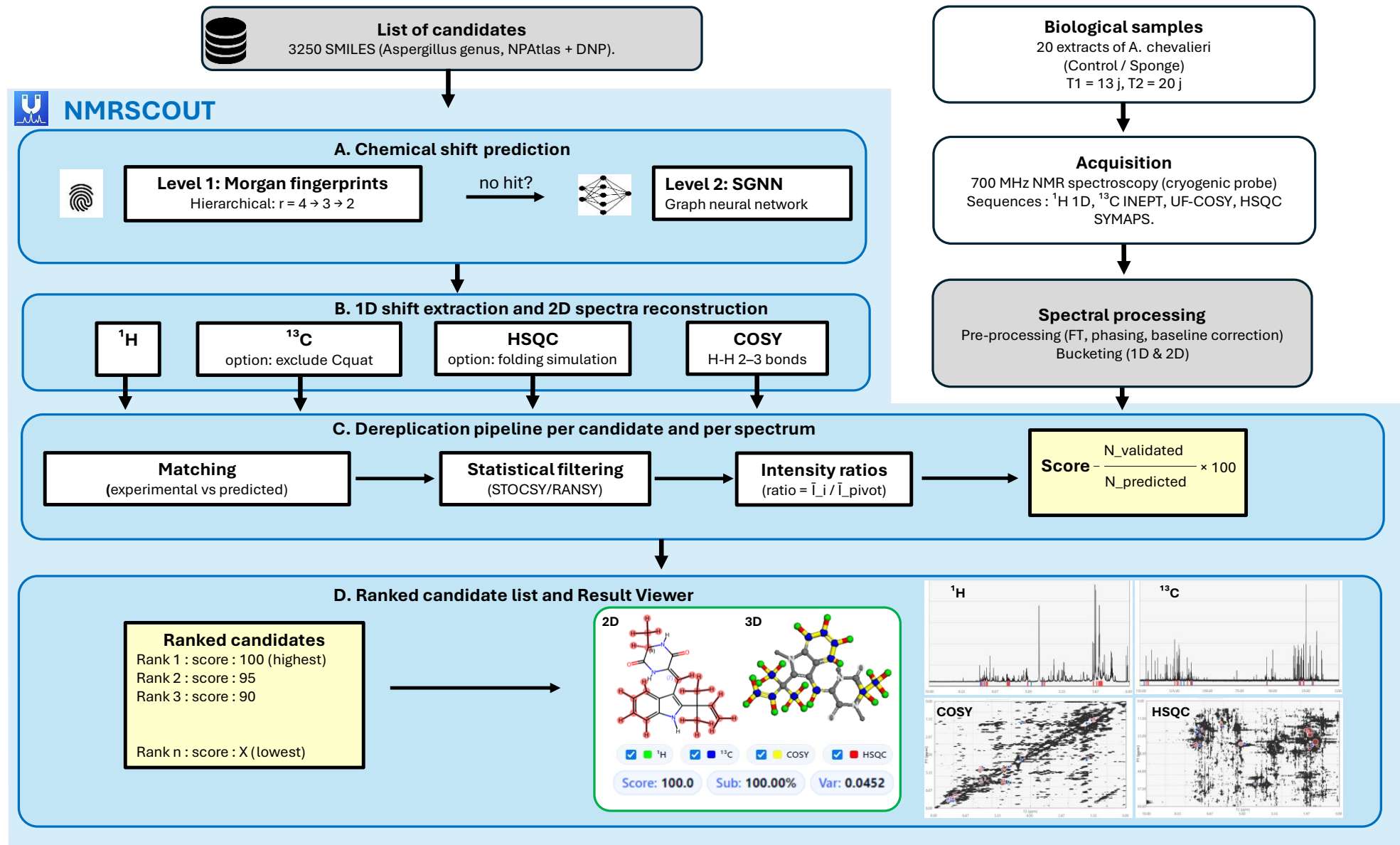
1D: ^1H , ^{13}C | 2D: COSY, HSQC

Automated annotation

Spectral matching + filters

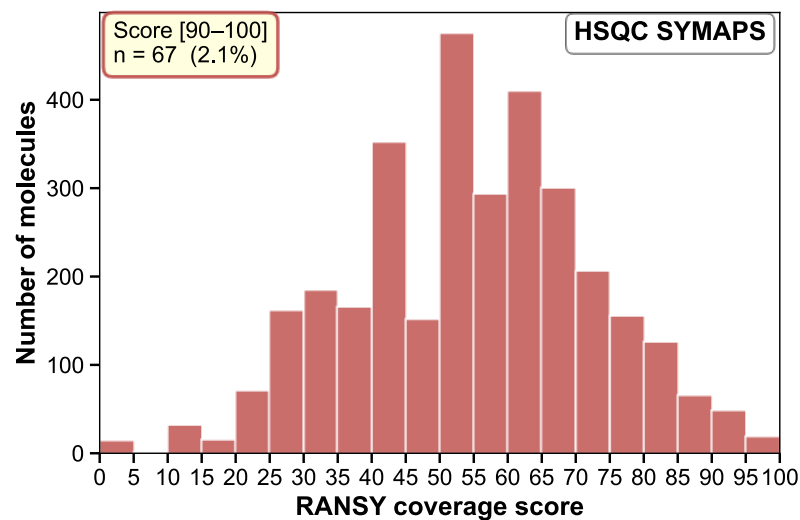
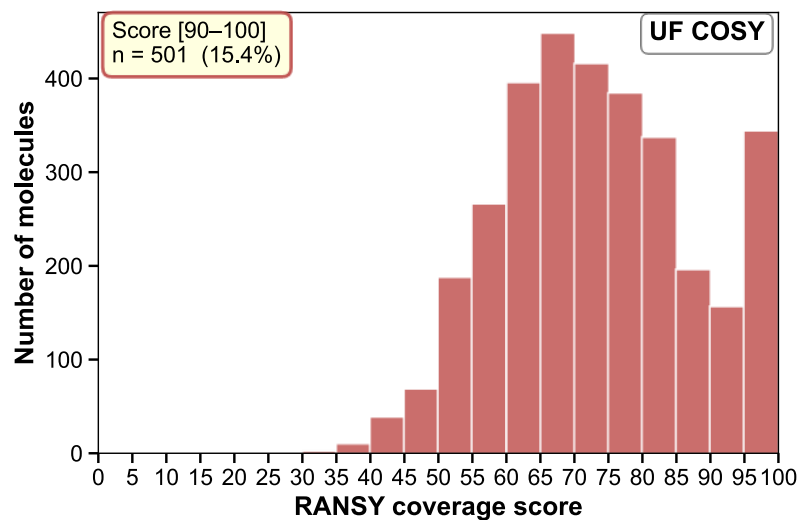
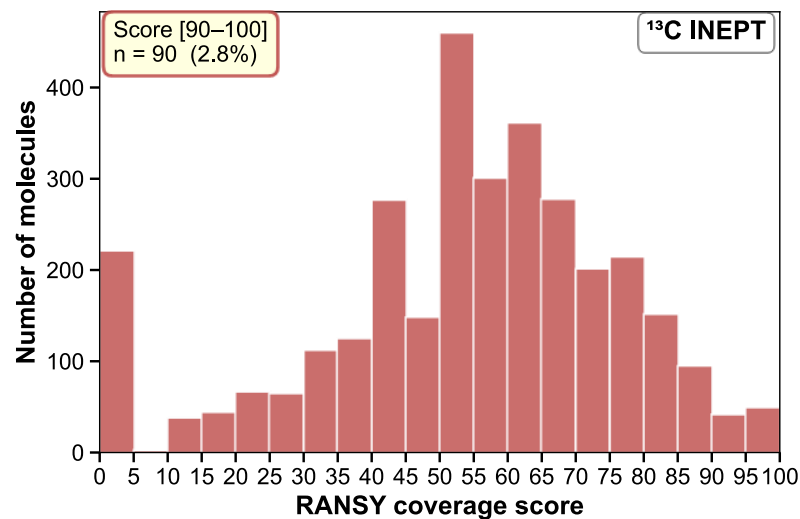
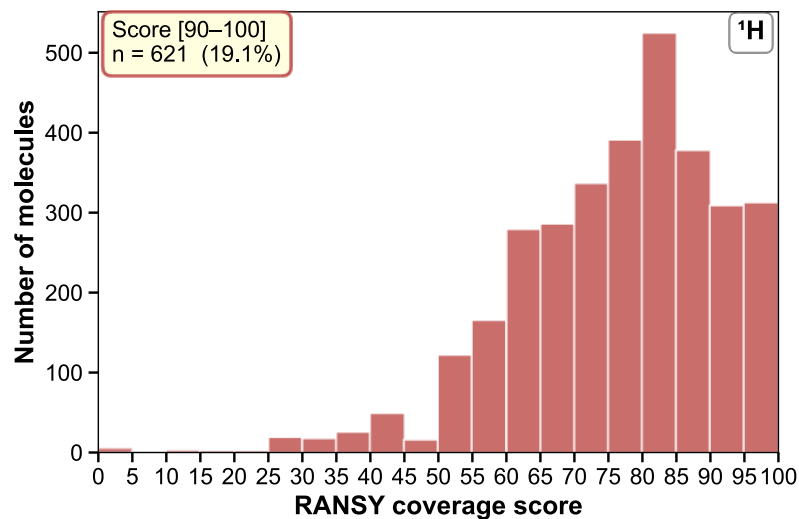


Pipeline overview



RESULTS

Distribution of RANSY coverage scores for each of the four NMR experiments

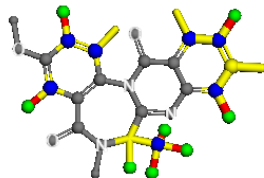
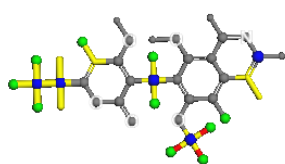
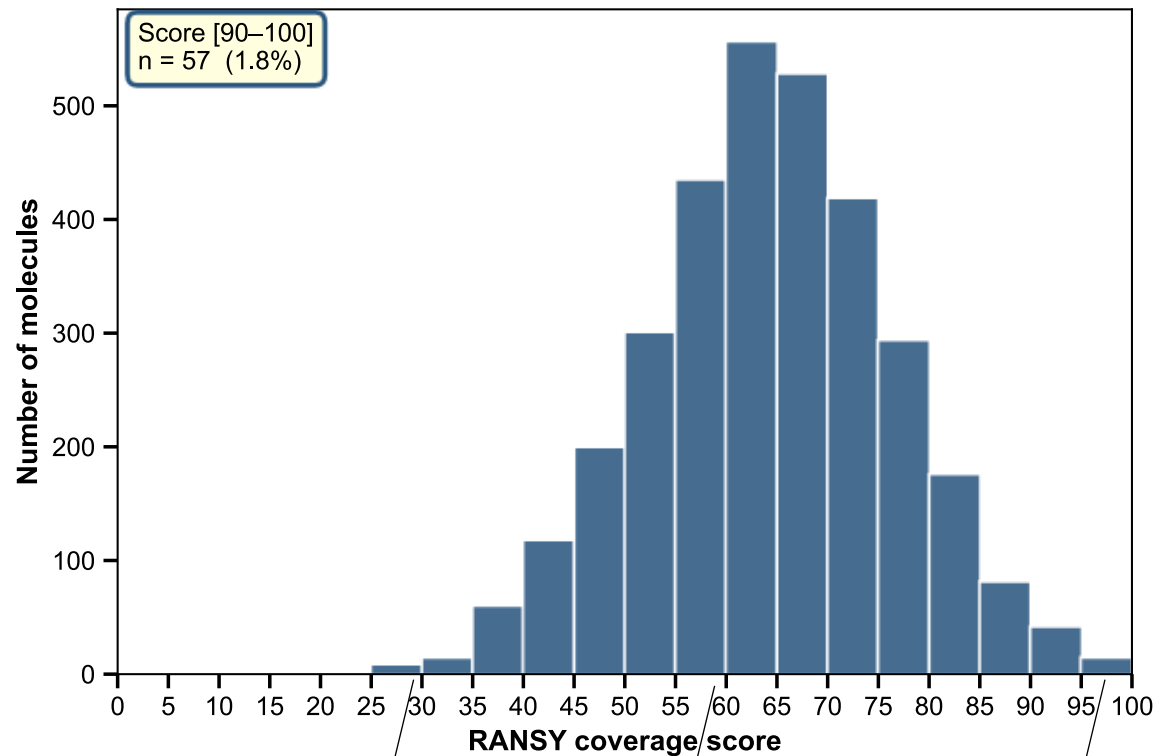


¹H and COSY: PERMISSIVE

¹³C and HSQC: SELECTIVE

RESULTS

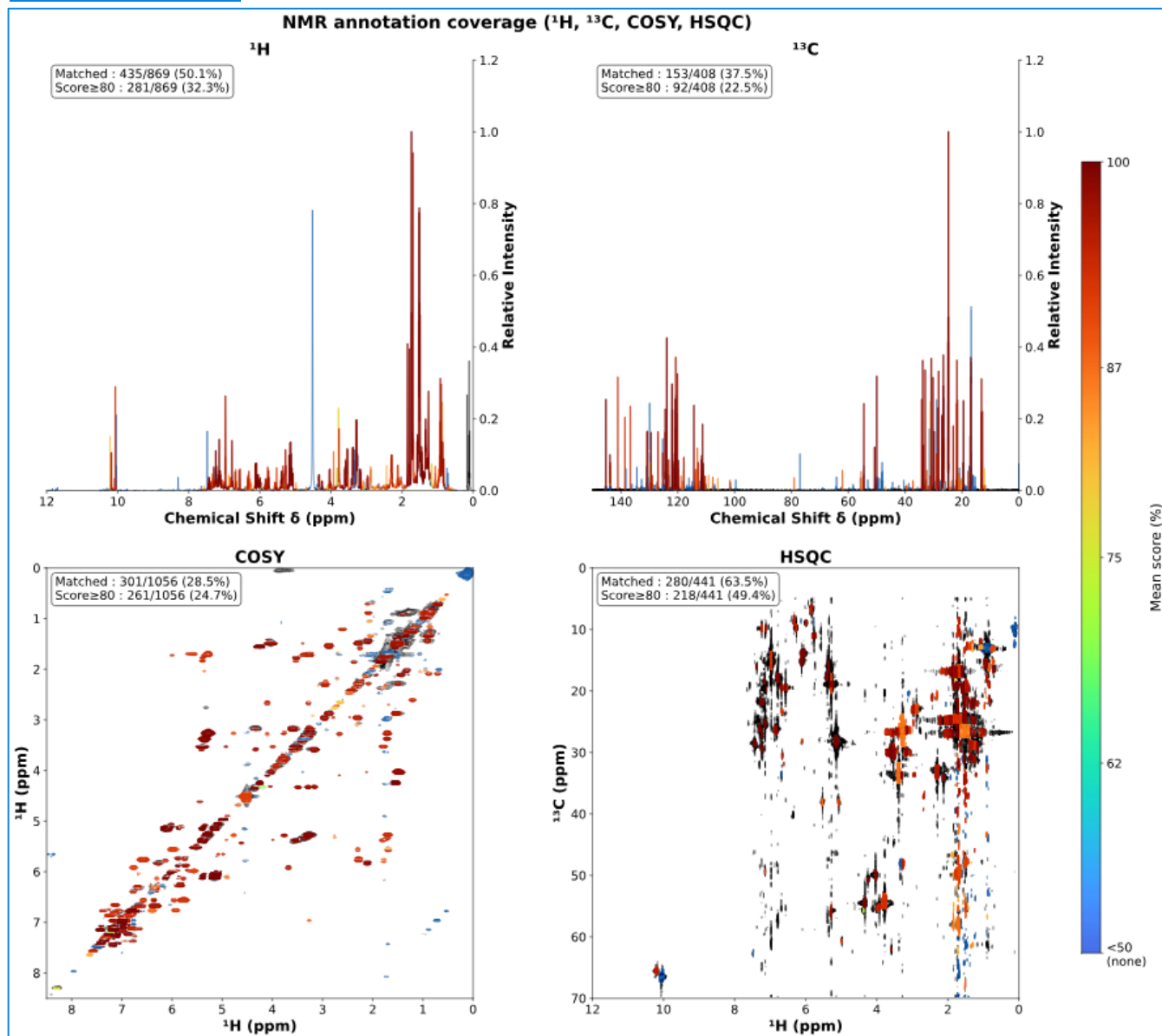
Distribution of mean scores of the four methods (^1H , ^{13}C INEPT, UF COSY et HSQC SYMAPS)



Only 1.8% of candidates
(n=57) reach the optimal
score [90–100]

Multi-experiment filtering for highly
selective structural identification

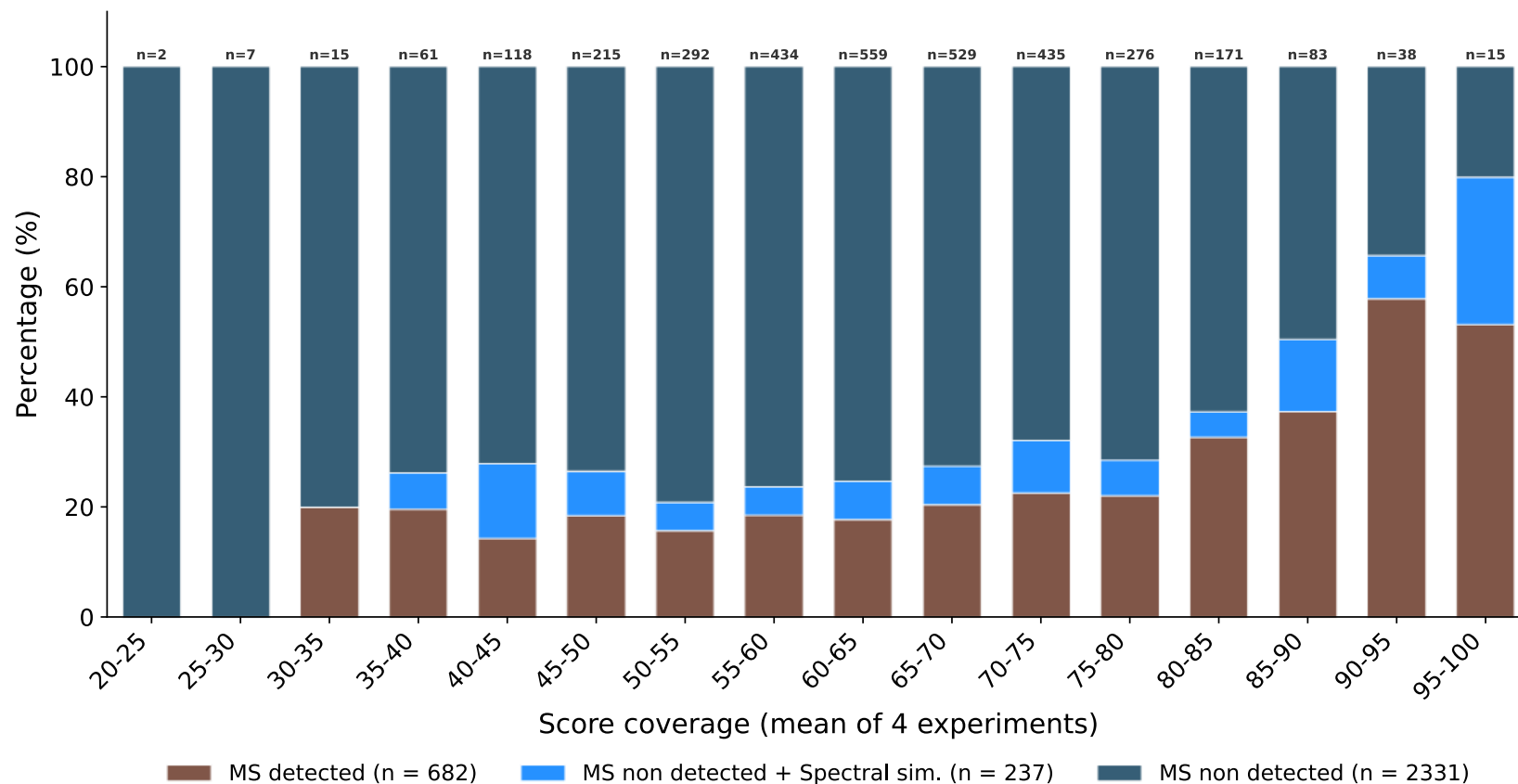
RESULTS



High-score candidates:
significant coverage of
experimental spectra

Unannotated signals: potentially
novel molecules, not covered
by the database

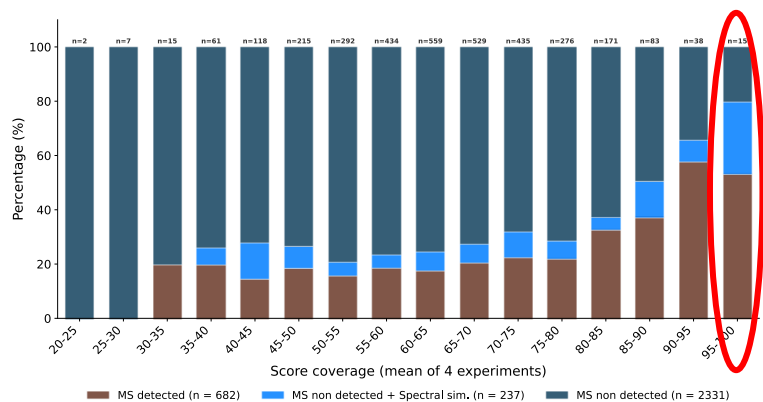
RESULTS



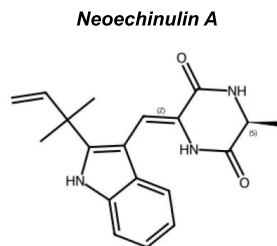
The proportion of MS detected candidates decreases with lower NMR annotation scores

Structural analogs at high scores: false positives due to spectral similarity with detected compounds

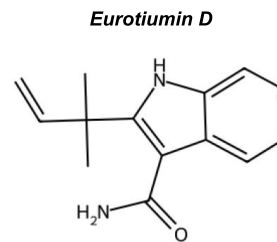
RESULTS



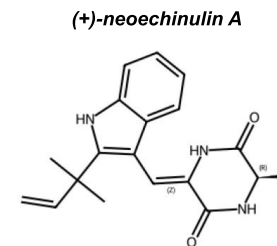
Top 15 Candidates with Highest Annotation Scores



NMR score: 100
m/z 324.1707
MS detected



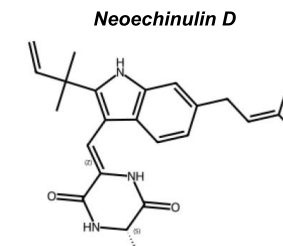
NMR score: 100
m/z: not detected
Spectral analog



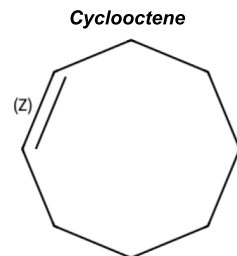
NMR score: 100
m/z 324.1707
MS detected



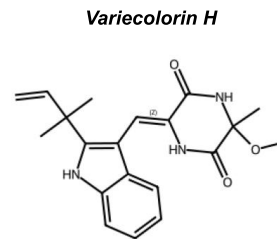
NMR score: 98
m/z 460.2959
MS detected



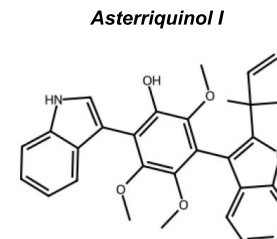
NMR score: 98
m/z: not detected
Spectral analog



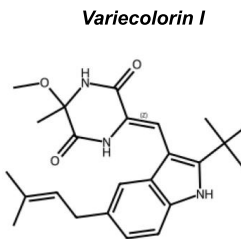
NMR score: 98
m/z: not detected
Not detected



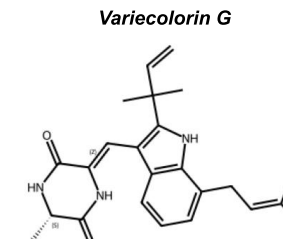
NMR score: 98
m/z 354.1812
MS detected



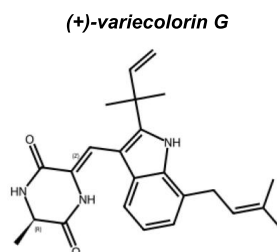
NMR score: 98
m/z: not detected
Not detected



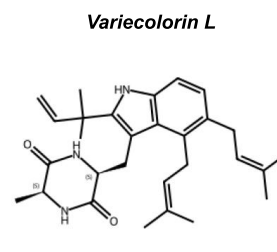
NMR score: 96
m/z 422.2438
MS detected



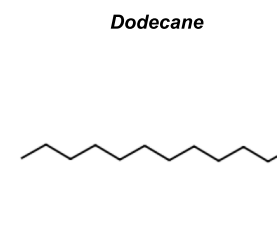
NMR score: 96
m/z: not detected
Spectral analog



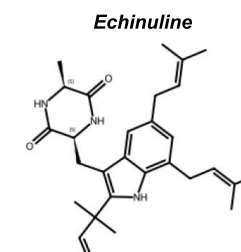
NMR score: 96
m/z: not detected
Spectral analog



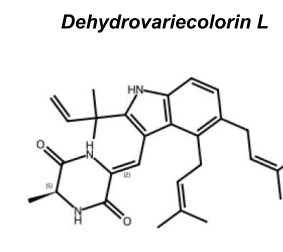
NMR score: 96
m/z 462.3115
MS detected



NMR score: 96
m/z: not detected
Not detected

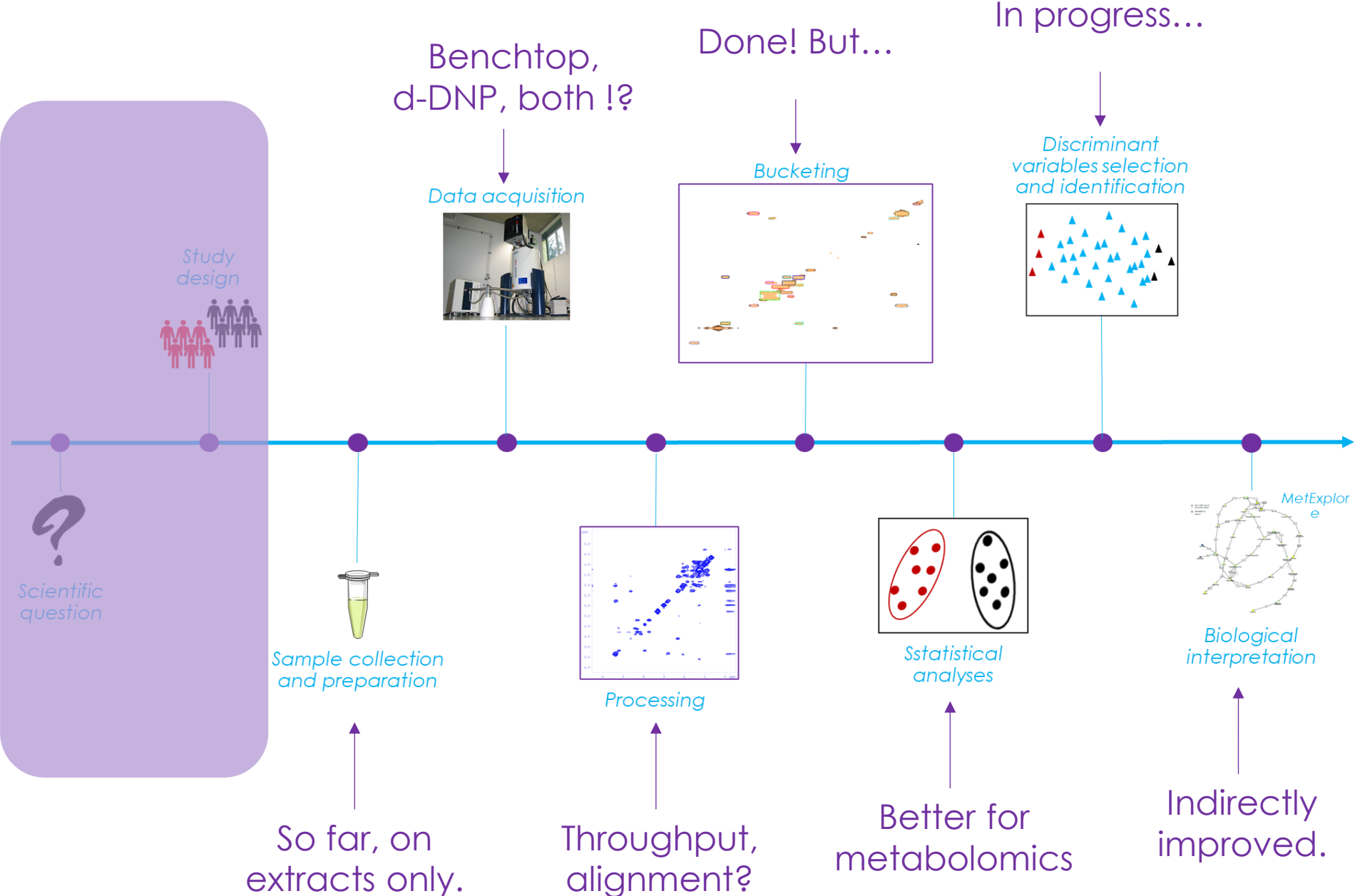


NMR score: 95
m/z 462.3115
MS detected



NMR score: 95
m/z 460.2959
MS detected

Conclusion & Perspectives



Acknowledgments

Equipe MIMM

Patrick Giraudeau
Jean-Nicolas Dumez
Virginie Silvestre
Estelle Martineau
Jonathan Farjon
Marine Piou
Victor Ribay
Pia Mayer
Benoît Charrier
Julien Gauvreau
Sumit Mishra
Martin Gualbert
Marie Delage
Jeremy Marchand
Arnab Day
Stéphane Guérin



Li Da Wie

Rafael Bruschweiler

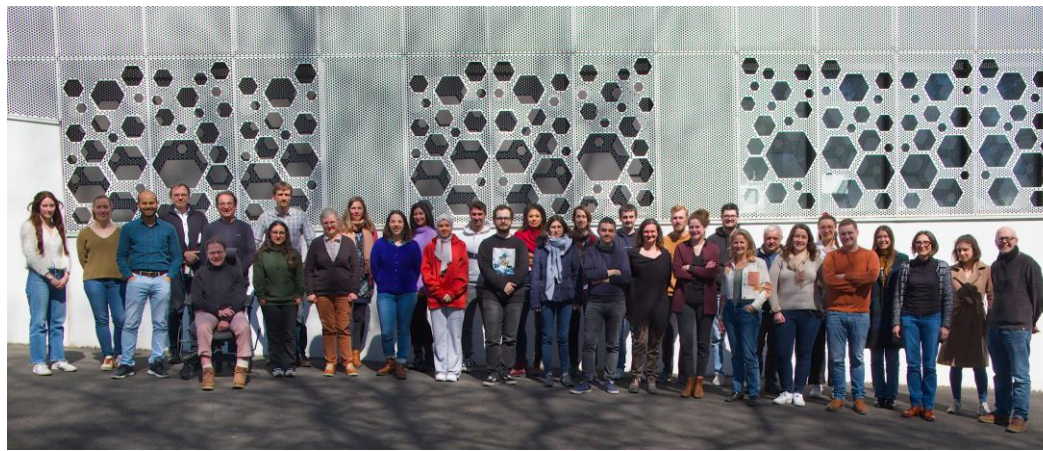


Gaud Dervilly

Maykel Hernández-Mesa

Axel Raux

Yann Guitton

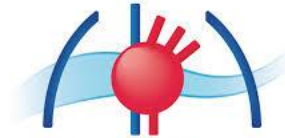


UNIVERSITÉ
DE GENÈVE

Julien Boccard



Claire Cannet



l'institut
du thorax

Mikaël Croyal
Chloé Cloteau
Bertrand Cariou
Samy Hadjadj



Equipe M3

Serigne Seck
Olivier Grovel
Samuel Bertrand
Aurore Michaud



Julien Guilbert
CANLET Cécile

TREMBLAY-FRANCO Marie

